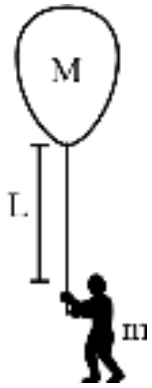
PHYSICS

1) The balloon of mass M, the light rope and the man of mass m shown in the figure is at rest and in equilibrium in the air. If the man starts climbing and reaches the top of the rope, then choose the correct statement



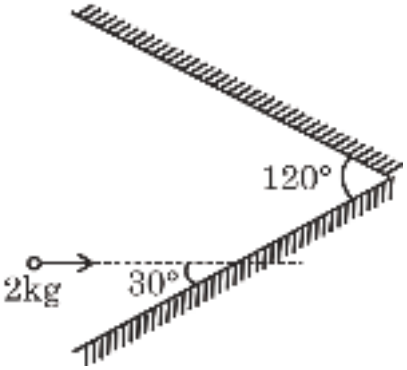
(1) Distance by which balloon descend is L

(2) Distance by which man ascend is $\frac{mL}{m+M}$

(3) Potential energy of man increases

(4) Change in gravitational potential energy of (man + balloon) will depend on their mass ratio.

2) A ball of mass 2kg moves with 10 m/s velocity in a smooth horizontal plane. It collides with one vertical smooth wall at an angle 60° from perpendicular to wall and after some time it collides to another wall as shown in figure. All collisions are elastic then find magnitude of net impulse provided by both walls?



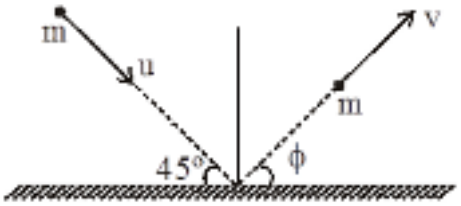
(1) 40 Ns

(2) 20 Ns

(3) 30 Ns

(4) Zero

3) A particle of mass m moving with a speed u strikes a smooth horizontal surface at an angle 45º the particle rebounds it an angle ϕ with speed v, if coefficient of restitution is $\frac{1}{\sqrt{3}}$, then angle ϕ is:-



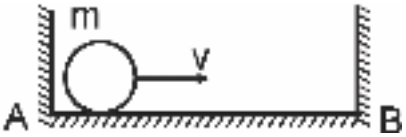
(1) 60°

(2) 30°

(3) 45°

(4) 53°

4) A small ball on a frictionless horizontal surface moves towards right with a velocity V. It collides with the wall and returns back and continues to and fro motion. If the average speed for first to and fro motion of the ball is $\left(\frac{2}{3}\right)$V, then the coefficient of restitution of impact is:



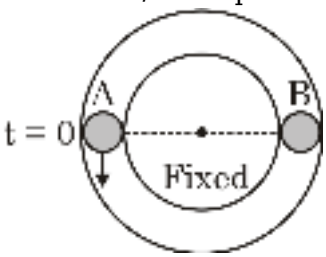
(1) 0.5

(2) 0.8

(3) 0.25

(4) 0.75

5) Particle 'A' moves with speed 10 m/s in a frictionless circular fixed horizontal pipe of radius 5 m and strikes with 'B' of double mass that of A. Coefficient of restitution is 1/2 and particle 'A' starts its journey at t = 0. The time at which second collision occurs is :-



(1) $\frac{\pi}{2}$s

(2) $\frac{2\pi}{3}$ s

(3) $\frac{5\pi}{2}$ s

(4) 4πs

6) A ball of mass m_1 moving with a velocity v collides head on elastically with another stationary ball of mass m_2. If after collision velocity of ball m_1 becomes $\frac{v}{3}$, then value of ratio $\frac{m_1}{m_2}$, is

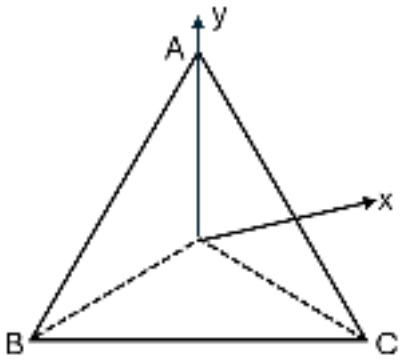
(1) 1

(2) 2

(3) 3

(4) 4

7) A uniform wire frame ABC is in the shape of an equilateral triangle in xy-plane. Select the correct statement if the centroid is the origin :-



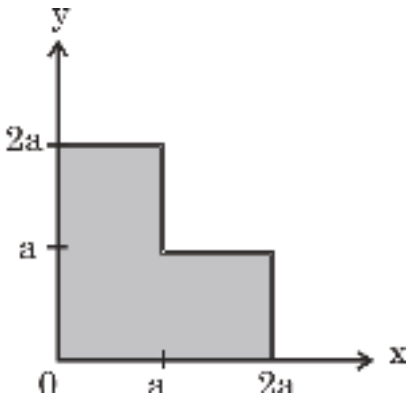
(1) If AB is removed, the centre of mass of the remaining figure is in second quadrant.

(2) If AC is removed, the centre of mass of the remaining figure is in first quadrant.

(3) If BC is removed, the centre of mass of the remaining figure is in fourth quadrant.

(4) If AB is removed, the centre of mass of the remaining figure is in fourth quadrant.

8) The figure shows a "L" shape object of uniform mass per unit area .



(A) x coordinate of center of mass is 5a/6
(B) y coordinate of center of mass is 5a/6
(C) x coordinate of center of mass is 2a/3
(D) y coordinate of center of mass is 3a/4

(1) A & D

(2) A & B

(3) C & B

(4) C & D

9) A two particle system of masses 2 kg and 3 kg present at (15,0) and (0, 20) in a x,y plane then its centre of mass doesn't lie on line :-

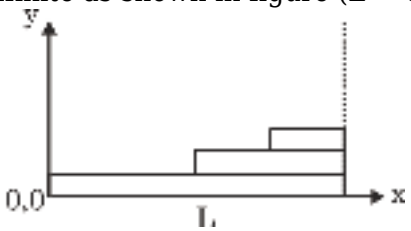
(1) y = 2x

(2) $\frac{x}{12} + \frac{y}{24} = 1$

(3) 3x – y = 6

(4) y = x

10) Infinite rods of uniform mass density and length L, L/2, L/4.... are placed one upon another upto infinite as shown in figure (L = 6). Find the x-coordinate of centre of mass.

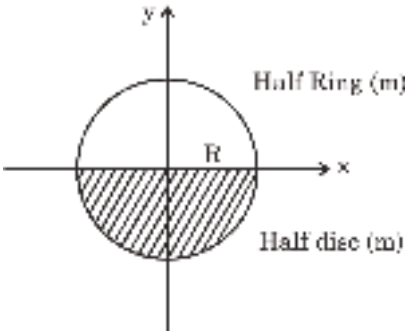


(1) 3 units

(2) 1 units

(3) 2 units

(4) 4 units



11) y-coordinate of COM of given system will be at :-

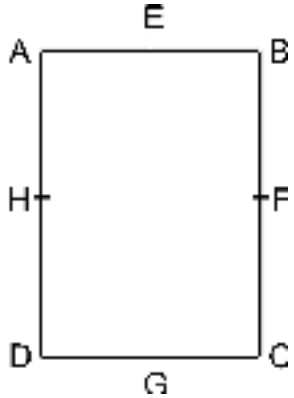
(1) $\frac{R}{3\pi}$

(2) $\frac{-R}{3\pi}$

(3) $\frac{2R}{3\pi}$

(4) $\frac{-2R}{3\pi}$

12) In a system of four thin uniform rods each of mass m as shown. If AE = EB, AH = HD and BC = 2 AB then choose the **CORRECT** statement :-



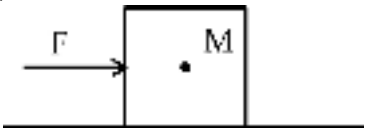
(1) Moment of inertia of the system is minimum about EG

(2) Moment of inertia of the system is maximum about HF

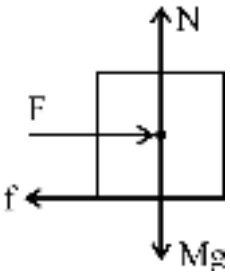
(3) Moment of inertia of the system is minimum about HF

(4) Moment of inertia of the system is maximum about EG

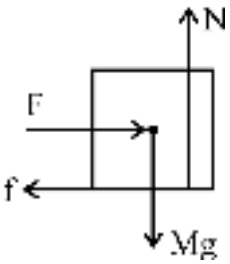
13) If the block shown in the figure is in equilibrium, under the action of an external force. Then choose the correct free body diagram of the block :-



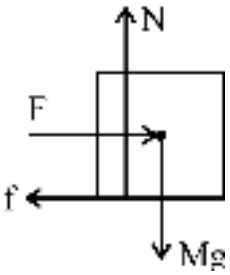
(1)



(2)

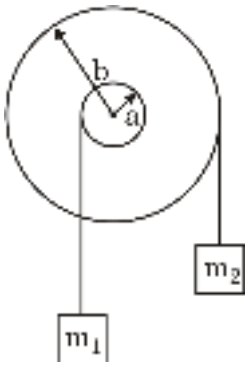


(3)



(4) None of these

14) For the wheel-and-axle system shown, which of the following expresses the condition required for the system to be in static equilibrium?



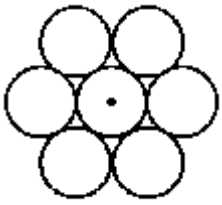
(1) $m_1 = m_2$

(2) $am_1 = bm_2$

(3) $am_2 = bm_1$

(4) $a^2m_1 = b^2m_2$

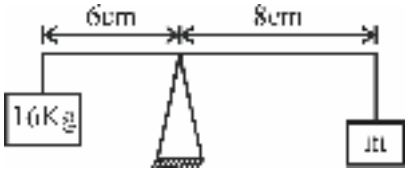
15) Seven pennies are arranged in a hexagonal, planar pattern so as to touch each neighbor, as shown in the figure below. Each penny is a uniform disk of mass m = 2kg and radius r = 1m. What is the moment of inertia of the system (in kg-m^2) of seven pennies about an axis that passes through



the center of the central penny and is normal to the plane of the pennies ?

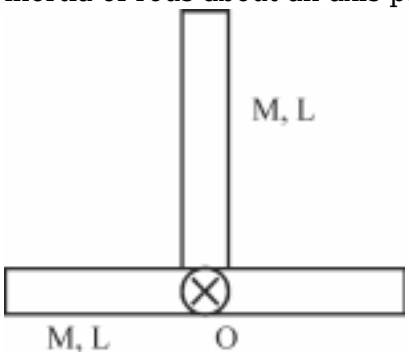
(1) 64 units
(2) 46 units
(3) 55 units
(4) 31 units

16) In an experiment with a beam balance on unknown mass m is balanced as shown in fig. Find the value of the unknown mass m (in Kg):



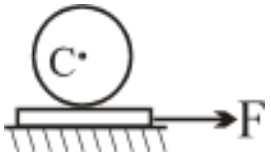
(1) 4 kg
(2) 16 kg
(3) 8 kg
(4) 12 kg

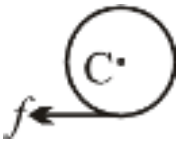
17) Two identical rods each of mass M and length L are kept according to figure. If the moment of inertia of rods about an axis passing through O and perpendicular to the plane of rods : -



(1) $\frac{ML^2}{3}$
(2) $\frac{5ML^2}{12}$
(3) $\frac{5ML^2}{3}$
(4) $\frac{ML^2}{12}$

18) A cylinder is placed on a rough plank which in turn is placed on a smooth surface. The plank is pulled with a constant force F. The friction force can be given by which of the following diagrams.



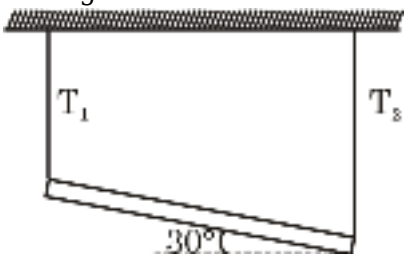
(1) 

(2) 

(3) 

(4) canot be interpreted

19) A uniform rod of mass m and length ℓ is connected with two ideal strings from ends of rod as shown in figure, if tensions in strings are T_1 & T_2 respectively then :-



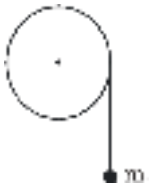
(1) $T_1 > \frac{mg}{2}$

(2) $T_2 > \frac{mg}{2}$

(3) $T_1 < \frac{mg}{2}$

(4) $T_1 = T_2 = \frac{mg}{2}$

20) A uniform solid cylinder of mass m can rotate freely about its fixed axis, which is horizontal. A particle of mass m_0 hangs from the end of a light string which is tightly wound round the cylinder as shown in figure. When system is allowed to move, acceleration with which the particle descends.



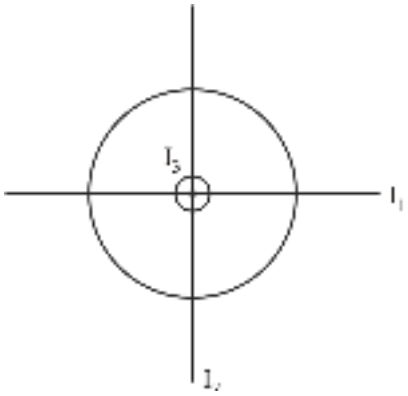
(1) $\frac{mg}{m + 2m_o}$

(2) $\frac{2m_o g}{m + 2m_o}$

(3) $\frac{m_o g}{m + m_o}$

(4) $\frac{2mg}{m + 2m_o}$

21) I_1 and I_2 are diametric axis and I_3 is axis perpendicular to plane of disc and passing through



centre of disc having mass m and radius R, $I_1 + I_2 + I_3$ will be :

(1) mR^2

(2) $\frac{mR^2}{2}$

(3) $\frac{mR^2}{4}$

(4) $2mR^2$

22) A solid sphere rolls down without slipping on a 30° inclined plane. If g = 10 m/s^2, the acceleration of the rolling sphere is:

(1) 5 $m\ s^{-2}$

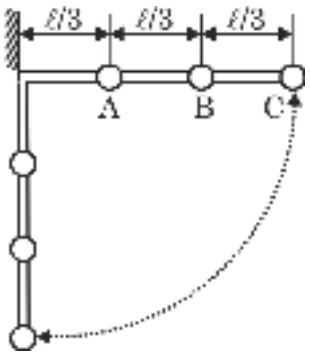
(2) $\frac{7}{25}\ m\ s^{-2}$

(3) $\frac{25}{7}\ m\ s^{-2}$

(4) $\frac{15}{7}\ m\ s^{-2}$

23)

A light rod carries three equal masses A, B and C as shown in figure. The velocity of B in vertical position of rod if it is released from horizontal position as shown in figure is :-



(1) $\sqrt{2g\ell}$

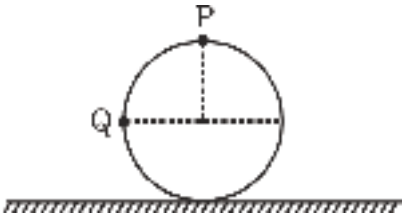
(2) $\sqrt{\frac{18g\ell}{7}}$

(3) $\sqrt{\frac{4g\ell}{3}}$

(4) $\sqrt{\frac{8g\ell}{7}}$

24)

A ball is pure rolling on the ground. The ratio of speed at P and speed at Q is :



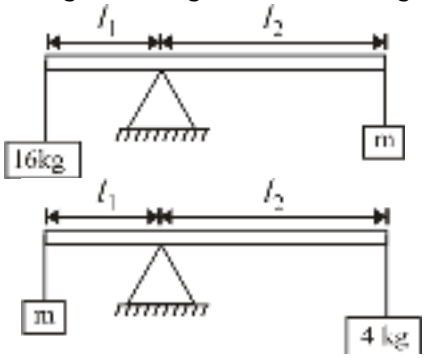
(1) $\sqrt{2}:1$

(2) 2 : 1

(3) $2\sqrt{2}:1$

(4) 1 : 2

25) In an experiment with a beam balance an unknown mass m is balanced by two known masses of 16 kg and 4 kg as shown in figure. Find the value of the unknown mass m in kg.



(1) 2 kg

(2) 4 kg

(3) 8 kg

(4) 12 kg

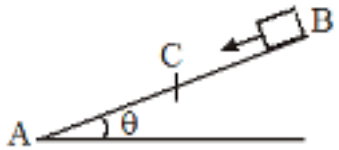
26) A stone tide to a string of length L is whirled in a vertical circle with the other end of the string at the centre. At a certain instant of time, the stone is at its lowest position and has a speed u. The magnitude of change in its velocity, as it reaches a position where the string is horizontal, is $\sqrt{x(u^2-gL)}$. The value of x is

(1) 3

(2) 2

(3) 1

(4) 5

27) 

A small block starts slipping down from a point B on an inclined plane AB, which is making an angle θ with the horizontal section BC is smooth and the remaining section CA is rough with a coefficient of friction m. It is found that the block comes to rest as it reaches the bottom (point A) of the inclined plane. If BC = 2AC, the coefficient of friction is given by : -

(1) μ = tanθ

(2) μ = 2tanθ

(3) μ = 3tanθ

(4) μ = 4tanθ

28) Body of mass m accelerates uniformly from rest to a speed v_0 in time t_0. The work done on the body till anytime t is :-

(1) $\frac{1}{2}mv_0^2\left(\frac{t^2}{t_0^2}\right)$

(2) $\frac{1}{2}mv_0^2\left(\frac{t_0}{t}\right)$

(3) $mv_0^2\left(\frac{t}{t_0}\right)$

(4) $mv_0^2\left(\frac{t}{t_0}\right)^3$

29) The potential energy of a 1 kg particle free to move along the x-axis is given by :-

$U(x) = \left(\frac{x^4}{4} - \frac{x^2}{2}\right)J$

The total mechanical energy of the particle is 2 J. Then the maximum speed (in m/s) is?

(1) $\frac{1}{\sqrt{2}}$ m/s

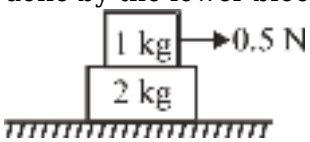
(2) $\sqrt{\frac{3}{2}}$ m/s

(3) $\frac{3}{2}$ m/s

(4) $\frac{3}{\sqrt{2}}$ m/s

30) A force of 0.5 N is applied on the upper block as shown in figure. The coefficient of static friction between the two blocks is 0.1 and that between the lower block and the surface is zero. The work

done by the lower block on the upper block for a displacement of 3 m of the upper block is :-



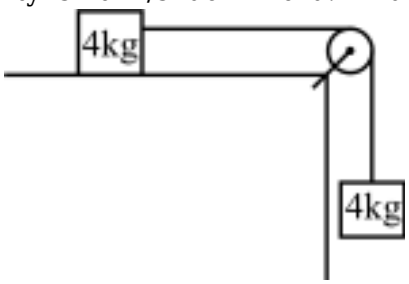
(1) 1 J

(2) –1 J

(3) 2 J

(4) –2 J

31) In the given arrangement, there is no friction any where. Pulley and string are ideal. Acceleration due to gravity is 10 m/s^2 downward. Find out total work done by gravity in 2 sec, if



system starts from rest.

(1) 100 J

(2) 400 J

(3) 200 J

(4) 800 J

32) A block slides down a smooth inclined plane when released from the top, while another falls freely from the same point :-

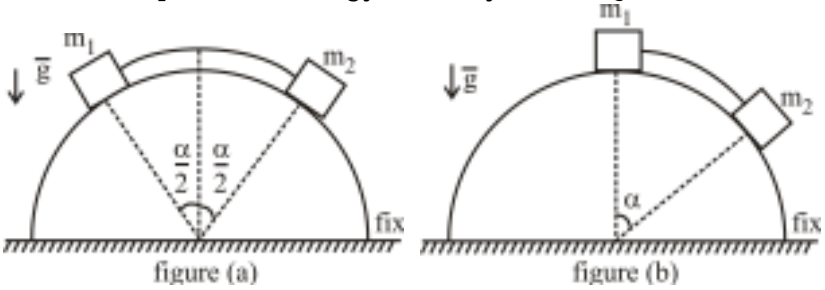
(1) Sliding block will reach the ground first with greater speed

(2) Freely falling block will reach the ground first with greater speed

(3) Both the blocks will reach the ground at the same time but with different speeds

(4) Both the blocks will reach the ground with same speed but the freely falling block first.

33) Two small body of mass m_2 = 6 kg and m_1 = 2 kg is connected by a massless thread and placed on a fixed smooth cylindrical surface of radius R = 0.5 m as shown in Figure (a). After the body is released from rest, they begin to move and attain the position shown in figure (b). Find the decrease in gravitational potential energy of the system (in Joule). Take α = 53°; cos 53° = 0.6, cos 26.5° = 0.9.



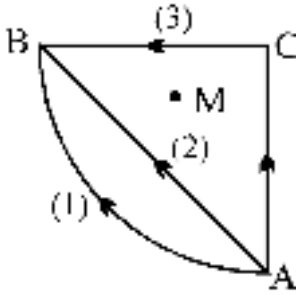
(1) 8 J

(2) 10 J

(3) 6 J

(4) 16 J

34) In a region of only gravitational field of mass 'M' a particle is shifted from A to B via three different paths in the figure. The work done in different paths are W_1, W_2, W_3 respectively then



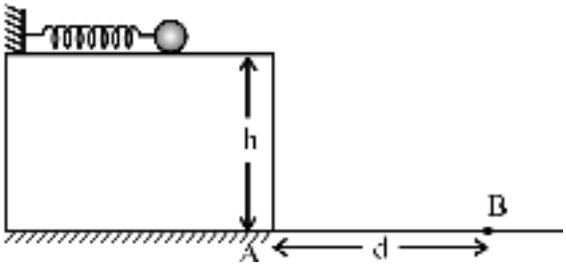
(1) $W_1 = W_2 = W_3$

(2) $W_1 = W_2 > W_3$

(3) $W_1 > W_2 > W_3$

(4) $W_1 < W_2 < W_3$

35) A ball of mass m is fired from a spring loaded gun that is mounted on a fixed table. If ball strikes to point B on ground, then find out initial compression of the spring. (Assume table is smooth and spring constant of spring is k) :



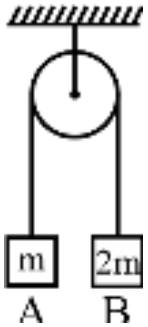
(1) $d\sqrt{\frac{mg}{2kh}}$

(2) $\frac{dmg}{2kh}$

(3) $\frac{d\sqrt{2h}}{mg}$

(4) $d\sqrt{\frac{2kh}{mg}}$

36) Block A & B are released from rest at t = 0, then power of tension force for block A at t = 2 sec is given by : -



(1) $\frac{9mg^2}{8}$

(2) $\frac{8mg^2}{9}$

(3) $\frac{4mg^2}{9}$

(4) $\frac{9mg^2}{4}$

37) In the given figure the bob of the pendulum is given an initial velocity u_0 in horizontal direction.



List-I (Nature of AB & u_0)		List-II (Maximum height attained by bob)	
(P)	AB → massless rod; $u_0 = \sqrt{3g\ell}$	(1)	$\frac{40}{27}\ell$
(Q)	AB → massless rod; $u_0 = \sqrt{4g\ell}$	(2)	$\frac{3\ell}{2}$
(R)	AB → massless string; $u_0 = \sqrt{5g\ell}$	(3)	2
(S)	AB → massless string; $u_0 = \sqrt{3g\ell}$	(4)	4

(1) P → 2 ; Q → 3 ; R → 3 ; S → 1

(2) P → 2 ; Q → 3 ; R → 4 ; S → 1

(3) P → 3 ; Q → 3 ; R → 3 ; S → 2

(4) P → 3 ; Q → 2 ; R → 4 ; S → 2

38) ***Assertion :*** If work done by the force depends on path then force is said to be conservative.
Reason : All the conservative forces are central force.

(1) Both **Assertion** and **Reason** are true but **Reason** is NOT the correct explanation of **Assertion**.

(2) **Assertion** is true but **Reason** is false.

(3) **Both Assertion** and **Reason** are false.

(4) Both **Assertion** and **Reason** are true and **Reason** is the correct explanation of **Assertion**.

39) A particle is moving in a circle of radius R = 2m, with an angular velocity $\omega = 4$ rad/sec and angular acceleration $\alpha = -2$ rad/sec^2. The total angular distance travelled by the particle in 3 seconds is given by : -

(1) 3 rad

(2) 4 rad

(3) 5 rad

(4) 6 rad

40) Angular displacement of an object moving on a circle of radius 2m is given by $\theta = \frac{\pi}{6}t + \frac{\pi}{6}t^2$ where θ is in radian and t is in s. For a time duration of t = 0 to t = 1s :

(A) Average angular velocity is $\frac{\pi}{3}$ rad/s

(B) Average angular velocity is $\frac{\pi}{6}$ rad/s

(C) Average velocity is $\frac{3\pi}{3}$ m/s

(D) Average velocity is 2 m/s

(1) A and C

(2) B and D

(3) A and D

(4) B and C

41) A curved road with radius of curvature 200 m is banked with an angle of banking equal to $\tan^{-1}(0.2)$. Now, if the traffic is at double the speed for which the road is designed, the minimum value of the frictional coefficient needed is : -

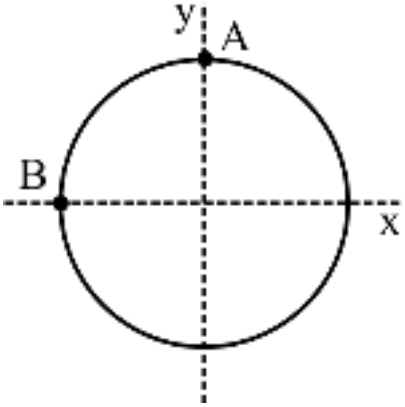
(1) $\frac{25}{29}$

(2) $\frac{15}{29}$

(3) $\frac{15}{21}$

(4) $\frac{25}{21}$

42) A toy racing car moves with constant speed around the circle shown below. When it is at point A its coordinates are x = 0, y = 3m and its velocity is $(6\text{m/s})\,\hat{i}$. When it is at point B its velocity and



acceleration are respectively :

(1) $-(6\text{m/s})\,\hat{j}$ and $\left(12\text{m/s}^2\right)\hat{i}$

(2) $(6\text{m/s})\,\hat{i}$ and $-\left(12\text{m/s}^2\right)\hat{i}$

(3) $(6\text{m/s})\,\hat{j}$ and $\left(12\text{m/s}^2\right)\hat{i}$

(4) $(6\text{m/s})\,\hat{i}$ and $\left(12\text{m/s}^2\right)\hat{j}$

43) A particle is moving with a constant angular acceleration of 16 rad./sec^2 in a circular path. At time t = 0 particle was at rest. Find the time at which the magnitudes of centripetal acceleration and tangential acceleration are equal :-

(1) 2 sec

(2) 4 sec

(3) 1 sec

(4) $\frac{1}{4}$ sec

44) The over-bridge of a river is in the form of a circular arc of radius of curvature r. If m is the combined mass of the motorcycle and the rider crossing the bridge at a speed v at highest point. The normal on the bridge at the highest point will be :

(1) $\frac{mv^2}{r}$

(2) mg

(3) $\frac{mv^2}{r} - mg$

(4) $mg - \frac{mv^2}{r}$

45) Which of the following statement(s) is/are true for a particle moving in a circle with a constant angular acceleration ?

(1) The magnitude of acceleration is constant.

(2) The acceleration vector is along the tangent to the circle.

(3) The velocity vector points along tangent to the circle.

(4) The velocity and acceleration vectors are always perpendicular to each other.

CHEMISTRY

1) The rate of a reaction is expressed in different ways as follows :

$$+\frac{1}{2}\frac{d[C]}{dt} = -\frac{1}{3}\frac{d[D]}{dt} = +\frac{1}{4}\frac{d[A]}{dt} = -\frac{d[B]}{dt}$$

The reaction is :

(1) $4A + B \rightarrow 2C + 3D$

(2) $B + 3D \rightarrow 4A + 2C$

(3) $A + B \rightarrow C + D$

(4) $B + D \rightarrow A + C$

2) A hypothetical reaction, $A_2 + B_2 \rightarrow 2AB$ follows the mechanism as given ahead.

$A_2 \rightleftharpoons A + A$ .....(fast)

$A + B_2 \rightarrow AB + B$ .....(slow)

$A + B \rightarrow AB$ .....(fast)

The order of the overall reaction is :

(1) 2

(2) 1

(3) $\frac{3}{2}$

(4) Zero

3) The unit of rate constant obeying the rate expression $r = k[A]^1[B]^{2/3}$ is :

(1) $mol^{-2/3}$ $litre^{2/3}$ $time^{-1}$

(2) $mol^{2/3}$ $litre^{-2/3}$ $time^{-1}$

(3) $mol^{-5/3}$ $litre^{5/3}$ $time^{-1}$

(4) none of these

4) For a first order reaction, if the time taken for completion of 50% of the reaction is t second, the time required for completion of 99.9% of the reaction is :

(1) 5 t

(2) 10 t

(3) 100 t

(4) 2 t

5) For the first order reaction with rate constant k, which expression gives the half-life period (Initial concentration = a) :

(1) $\frac{\ell n2}{k}$

(2) $\frac{1}{ka}$

(3) $\frac{0.693}{2k}$

(4) $\frac{3}{2ka^2}$

6) Half-lives of first order and zeroth order reactions are same. Ratio of rates at the start of reaction is :

(1) 0.693

(2) 2 × 0.693

(3) $\frac{1}{0.693}$

(4) $\frac{2}{0.693}$

7) The rate constant is numerically the same for three reaction of first, second and third order respectively. Which one is true for the three of reaction, if concentration of reactant is greater than 1M?

(1) $r_1 > r_2 > r_3$

(2) $r_1 = r_2 = r_3$

(3) $r_1 < r_2 < r_3$

(4) All of these

8) An elementary reaction between A and B is a second order reaction. Which of the following rate equations must be correct ?

(1) $r = k[A]^2 [B]^1$

(2) $r = k[A]^{3/2} [B]^{1/2}$

(3) $r = k[A]^2 [B]^2$

(4) $r = k[A][B]$

9) If the rate of gaseous reactions is independent of pressure, the order of reaction is :

(1) 0

(2) 1

(3) 2

(4) 3

10) The rate constant of a reaction becomes equal to the pre-exponential factor when :

(1) the absolute temperature is infinity

(2) the temperature in celsius is zero

(3) the absolute temperature is zero

(4) the activation energy is infinity

11) Rate constant depends on :

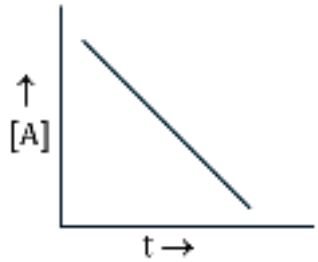
(1) Temperature

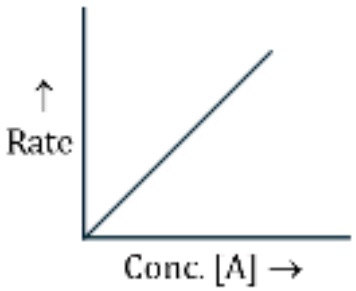
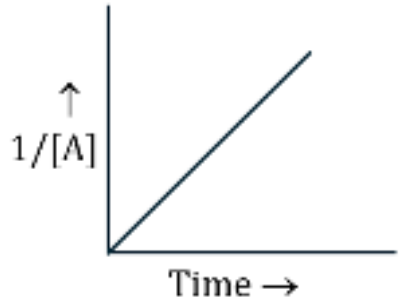
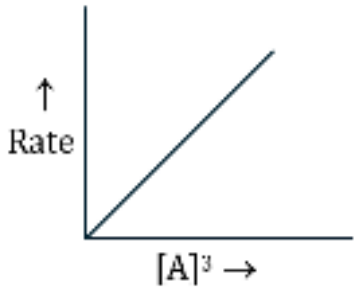
(2) Concentration

(3) Pressure

(4) All of these

12) **Match the column-I with Column-II :**

Column I (Head)	**Column II (Order)**
 (A)	(p) 1

(B) 	(q) 2
(C) 	(r) 3
(D) 	(s) 0

(1) (A) → (p); (B) → (s); (C) → (r); (D) → (q)

(2) (A) → (s); (B) → (p); (C) → (q); (D) → (r)

(3) (A) → (p); (B) → (q); (C) → (r); (D) → (s)

(4) (A) → (r); (B) → (q); (C) → (p); (D) → (q)

13) Match the half-lives in Column-I with the orders in Column-II :

Column I (Half-life)	**Column II (Order)**
(A) $t_{1/2}$ = constant	(p) First order
(B) $t_{1/2} \propto a$	(q) Second order
(C) $t_{1/2} \propto \frac{1}{a}$	(r) Zero order

where, a = Initial concentration of the reactant

(1) A → p; B → r; C → q

(2) A → q; B → p; C → r

(3) A → r; B → q; C → p

(4) A → p; B → q; C → r

14) The reaction of A_2 and B_2 follows the equation

$A_2(g) + B_2(g) \rightarrow 2AB(g)$

The following data were observed

	$[A_2]_0$	$[B_2]_0$	**Initial rate of formation AB(g) (in Ms^{-1})**
(1)	0.10	0.10	2.5×10^{-4}
(2)	0.20	0.10	5×10^{-4}
(3)	0.20	0.20	10×10^{-4}

The value of order of the reaction is :

(1) 2

(2) 3

(3) 4

(4) 0

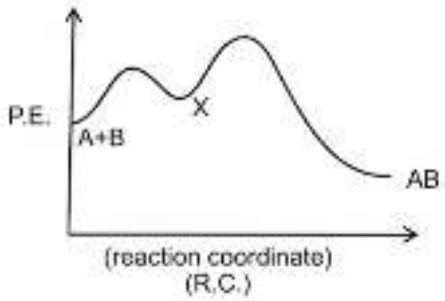
15) For an exothermic chemical process occurring in two steps as follows

(i) A + B → X (slow)

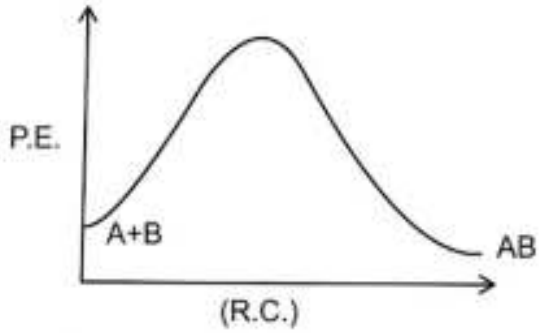
(ii) X → AB (fast)

The process of reaction can be best described by :

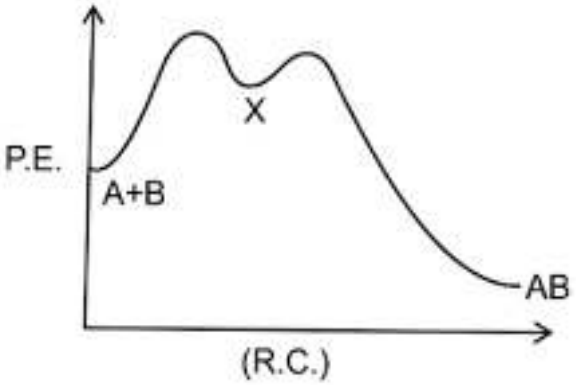
(1)

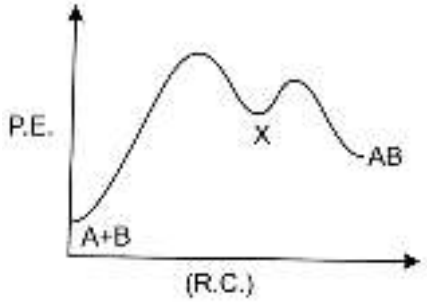


(2)



(3)



(4) 

16) Match the Column-I with Column-II :

Column-I (Solution)	**Column-II (van't Hoff factor assuming complete dissociation/association)**
(A) $Ca_3(PO_4)_2$(aq.)	(p) i > 1
(B) $Al_2(SO_4)_3$(aq.)	(q) i = 5
(C) Urea	(r) i = 1
(D) CH_3COOH in benzene	(s) i < 1

(1) (A) → (p, s); (B) → (r); (C) → (q, r); (D) → (p)

(2) (A) → (p, q); (B) → (p, q); (C) → (r); (D) → (s)

(3) (A) → (p); (B) → (q); (C) → (s); (D) → (r)

(4) (A) → (s); (B) → (q); (C) → (r); (D) → (p)

17) Freezing point of an aqueous solution is –0.186°C.
Elevation of boiling point of the same solution is ..... if K_b = 0.512 K molality^{-1} and K_f = 1.86 K molality^{-1}:

(1) 0.186°C

(2) 0.0512°C

(3) 0.092°C

(4) 0.237°C

18) The van't Hoff factor (i) for a dilute aqueous solution of the strong electrolyte barium hydroxide is:

(1) 0

(2) 1

(3) 2

(4) 3

19) A 5% (w/v) solution of cane sugar (molecular weight = 342) is isotonic with 1% (w/v) solution of a non-electrolyte substance X. The molecular weight of X is:

(1) 34.2

(2) 68.4

(3) 136.8
(4) 171.2

20) 0.5 M aqueous solution of glucose is isotonic with:

(1) 0.5 M KCl solution
(2) 0.5 M $CaCl_2$ solution
(3) 0.5 M urea solution
(4) 1 M solution of sucrose

21) Which of the following is dependent on temperature?

(1) Molarity
(2) Mole fraction
(3) Weight percentage
(4) Molality

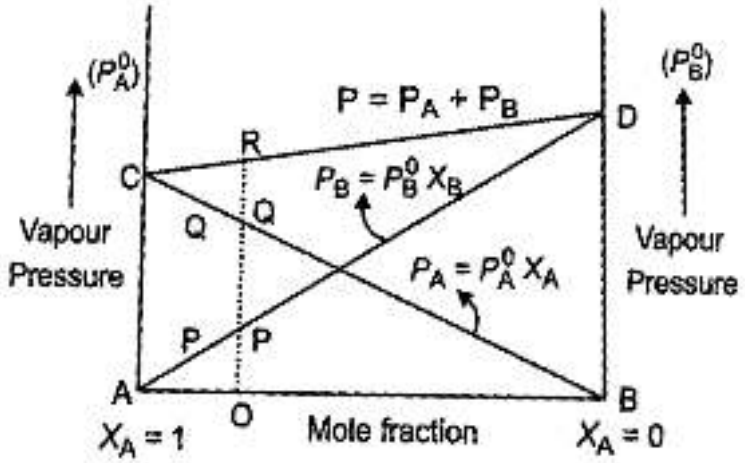
22) Normality of 10% (w/v) H_2SO_4 solution nearly :

(1) 0.1
(2) 0.2
(3) 0.5
(4) 2

23) 5 N H_2SO_4 was diluted from 1 litre to 10 litres. Normality of the solution obtained is:

(1) 10 N
(2) 5 N
(3) 1 N
(4) 0.5 N

24) Two volatile liquids A and B form ideal solution.
Considering the following vapour-pressure composition graph.



OR will be equal to :

(1) OQ + QR
(2) OQ + PQ
(3) OP + OQ
(4) OP + PR

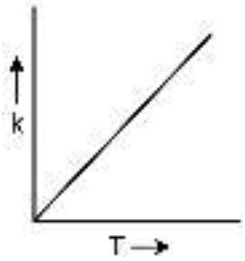
25) Which one is not equal to zero for an ideal solution:

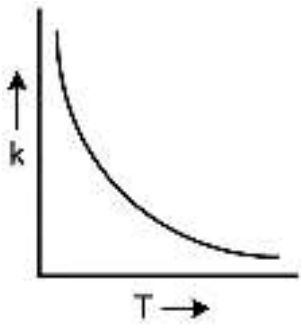
(1) ΔS_{mix}
(2) ΔV_{mix}
(3) ΔH_{mix}
(4) All of these

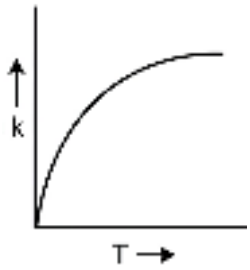
26) For an ideal solution of two components A and B, which of the following is true?

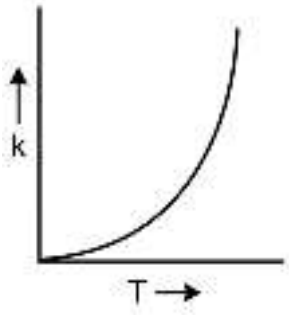
(1) $\Delta H_{mixing} > 0$ (zero).
(2) A–B interaction is stronger than A–A and B–B interactions.
(3) A–A, B–B and A–B interactions are identical.
(4) $\Delta H_{mixing} < 0$ (zero)

27) The rate constant of most of the reactions increases with increase in temperature. According to the Arrhenius equation, $k = Ae^{-E_a/RT}$. The curve of rate constant 'k' against temperature 'T' will be :

(1) 

(2) 

(3) 

(4) 

28) The rate constant of a reaction is 2.1×10^{-2} mol^{-2} $litre^{2}$ min^{-1}. The order of reaction is:

(1) 1
(2) 2
(3) 3
(4) zero

29) The chemical reaction, $2O_3 \rightarrow 3O_2$ proceeds as follows:
$O_3 \rightleftharpoons O_2 + O$ (fast)
$O + O_3 \rightarrow 2O_2$ (slow)
The rate law expression should be:

(1) $r = k[O_3]^2$
(2) $r = k[O_3][O_2]$
(3) $r = [O_3][O_2]^2$
(4) $r = k[O_3]^2[O_2]^{-1}$

30) The temperature coefficient for most of the reactions lies between:

(1) 1 and 3
(2) 2 and 3
(3) 1 and 4
(4) 2 and 4

31) The reaction $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$ is of first order. If volume of reaction vessel is reduced to $\frac{1}{3}$, the rate of reaction would be :

(1) $\frac{1}{3}$ times
(2) $\frac{2}{3}$ times
(3) 3 times
(4) 6 times

32) If in a 1st order reaction 75% of reaction complete in 4 hrs. Then how much time is required to complete 87.5% of reaction?

(1) 32 h

(2) 6 h

(3) 8 h

(4) 16 h

33) When initial concentration of a reactant is doubled in a reaction, its half-life period is not affected. The order of the reaction is :

(1) first

(2) second

(3) more than zero but less than first

(4) zero

34) Slope of the straight line obtained by plotting $\log_{10}$ k against 1/T represents which term?

(1) $-E_a$

(2) $-2.303\ E_a / R$

(3) $-E_a / (2.303\ R)$

(4) $-E_a / R$

35) If 18 g of glucose is present in 1000 g of solvent, the solution is said to be:

(1) 1 molar

(2) 0.1 molar

(3) 0.5

(4) 0.1 molal

36) The depression in f.pt. of 0.01 m aqueous solution of urea, sodium chloride and sodium sulphate is in the ratio:

(1) 1 : 1 : 1

(2) 1 : 2 : 4

(3) 1 : 2 : 3

(4) 2 : 2 : 3

37) Vapour pressure of a solution containing non-volatile solute is:

(1) inversely proportional to the mole fraction of the solute

(2) directly proportional to the mole fraction of the solvent

(3) inversely proportional to the mole fraction of the solvent

(4) directly proportional to the mole fraction of solute

38) Value of Henry's constant (K_H)__.

(1) decreases with increase in temperature

(2) increases with increase in temperature

(3) remains constant
(4) first increase then decreases

39) If molality of the dilute solutions is doubled, the value of molal depression constant (K_f) will be:

(1) halved
(2) tripled
(3) unchanged
(4) doubled

40) The mixture of two liquids boils without changing its composition is called:

(1) zeotropic liquid mixture
(2) binary liquid mixture
(3) azeotropic liquid mixture
(4) stable structure complex

41) What is the freezing point of a solution that contains 10.0 g of a glucose ($C_6H_{12}O_6$) in 100 g of H_2O ?
K_f = 1.86° C/m.

(1) –0.186º C
(2) +0.186º C
(3) –0.10º C
(4) –1.03º C

42) Which of the following is not a non-electrolyte?

(1) Acetic acid
(2) Glucose
(3) Ethanol
(4) Urea

43) Pure water can be obtained from sea water by:

(1) centrifugation
(2) plasmolysis
(3) reverse osmosis
(4) sedimentation

44) What is the van't Hoff factor for sodium phosphate?

(1) 1
(2) 2
(3) 3
(4) 4

45) Van't Hoff factor more than unity indicates that the solute in solution has:

(1) associated

(2) dissociated

(3) both

(4) cannot say anything

BIOLOGY

1) How many of the following characters are exhibited by living organism.
(I) All living organisms grow.
(II) Reproduction is a characteristics of living organism.
(III) Cellular organisation of the body is defining feature of life form.
(IV) All living organisms shows metabolism.
(V) Living organism shows consciousness.
(VI) All living organisms are linked to one another by sharing of the common genetic material, but to varying degrees.

(1) 4

(2) 5

(3) 6

(4) 3

2) Which of the character is not considered as defining feature of living beings ?

(1) Extrinsic growth

(2) Metabolism

(3) Intrinsic growth

(4) Cellular organization

3) In both plants and animals the most convenient method of nomenclature system is :-

(1) Polynomial nomenclature

(2) Binomial nomenclature

(3) Trinomial nomenclature

(4) More than one option is correct

4) Which of the following set of organisms do not reproduce :-

(1) Bacteria & Amoeba

(2) Mango & sterile worker bees

(3) Planaria & fungi

(4) Infertile human couples and mules

5) Consider the following statements (A-D) :-
(A) Biological names are Latinised or derived from Latin irrespective of their origin.

(B) Animals, mammals and dogs are different taxa at the different level
(C) By using artificial system of classification, we can get evolutionary relations in organisms.
(D) Bacteria cannot be considered as biological species according to the biological concept of species
Which of the above statements(s) are correct?

(1) A, B & D
(2) B, C & D
(3) B & C only
(4) A & B only

6) Which of the following is the smallest fundamental unit of taxonomy ?

(1) Order
(2) Genus
(3) Species
(4) Sub-species

7) Biological concept of species is based on :-

(1) Morphology
(2) Interbreeding
(3) Physiology
(4) All

8) What is the similarity between *Panthera* and *Solanum*?

(1) Both are group of plants
(2) Both are families
(3) Both are genus
(4) Both are order

9)

Study the given statements carefully and give the answer :-
(A) In lower taxa similarities are more
(B) Families are characterised on the basis of both vegetative and reproductive features of plants species.
(C) Felidae and canidae are the families of cat and dog respectively
(D) Determining relations in higher taxa is more difficult in comparison to lower taxa.

How many statements are correct from them
(1) One
(2) Two
(3) Three
(4) Four

10) The following list contains different organisms. To how many different genera these organisms belong ?
Tiger, Leopard, Cat, Lion, Mango, Wheat, Rice, Brinjal, Potato, Makoi, Housefly

(1) Eight
(2) Seven
(3) Nine
(4) Six

11) Match the column :

	Column-I		**Column-II**
(a)	Family	(i)	*tuberosum*
(b)	Kingdom	(ii)	Polymoniales
(c)	Order	(iii)	*Solanum*
(d)	Specific epithet	(iv)	Plantae
(e)	Genus	(v)	Solanaceae

(1) a-iv, b-v, c-ii, d-i, e-iii
(2) a-v, b-iv, c-ii, d-i, e-iii
(3) a-v, b-iv, c-i, d-iii, e-ii
(4) a-i, b-ii, c-iii, d-iv, e-v

12) Family of mango is ___

(1) Anacardiaceae
(2) Sapindales
(3) *Triticum*
(4) Muscidae

13) Potato, Brinjal, Makoi, *Petunia* and *Datura* have some common characters. So they are placed in the same :-

(1) Species
(2) Genus
(3) Tribe
(4) Family

14) Cats belong to _____ family whereas dogs belong to _____ family :-

(1) Felidae, Canidae
(2) Canidae, Felidae
(3) Muscidae, Felidae
(4) Muscidae, Canidae

15) How many of following statements are **correct**?
(i) Ecological information of organisms is one of the basis of modern taxonomic studies.
(ii) Word systematics is derived from *greek* word *systema*.
(iii) Category denotes rank and each rank is a unit of classification.
(iv) Earliest classifications were based on external morphology.

(1) 3
(2) 2
(3) 4
(4) 1

16) Which of the following forms motile colony ?

(1) *Volvox*
(2) *Hydrodictyon*
(3) *Porphyra*
(4) *Polysiphonia*

17) Algae are mainly classified on the basis of :

(1) Nature of stored food material in cell
(2) Structure organisation of thallus
(3) Chemical composition of cell wall
(4) Pigments present in cell

18) Which one of the following shows isogamy with non-flagellated gametes?

(1) *Sargassum*
(2) *Ectocarpus*
(3) *Ulothrix*
(4) *Spirogyra*

19) Sex organs of algae are usualy :

(1) Multicellular
(2) Unicellular
(3) Unicellular and Jacketed
(4) Multicellular and Jacketed

20) Read the following statements and select incorrect only :-
(a) *Ulothrix* and *Volvox* are filamentous algae.
(b) A few algae of marine forms such as kelps, form massive plant body.
(c) *Chlamydomonas* is microscopic, unicellular algae.
(d) *Porphyra, Laminaria* and *Spirogyra* are commonly used as food.

(1) b and d
(2) a and d

(3) a and b

(4) b and c

21) A protonema is _____,

(1) A structure in pteridophytes formed before the thallus develops.

(2) A sporophytic free living structure formed in pteridophytes.

(3) A creeping, green, filamentous and gametophytic structure produced in bryophytes.

(4) A primitive structure formed after fertilization in pteridophytes.

22) **Assertion (A):** Bryophytes are also called amphibians of the plant kingdom.
Reason (R): Bryophytes play an important role in plant succession on bare rocks/soil.

(1) (A) is correct but (R) is not correct.

(2) (A) is not correct but (R) is correct.

(3) Both (A) and (R) are correct and (R) is the correct explanation of (A).

(4) Both (A) and (R) are correct but (R) is not the correct explanation of (A).

23) From evolutionary point of view, retention of the female gametophyte with developing young embryo on the parent sporophyte for some time, is first observed in :

(1) Liverworts

(2) Mosses

(3) Pteridophytes

(4) Gymnosperms

24) Genera like *Selaginella* and *Salvinia* produce two kinds of spores. Such plants are known as :

(1) Homosorus

(2) Heterosorus

(3) Homosporous

(4) Heterosporous

25) A fern is different from moss in terms of having :-

(1) Dependent gametophyte

(2) Vascular tissue

(3) Swimming antherozoids

(4) Presence of alternation of generation

26)

Floridean starch is reserve food in :

(1) Phaeophyceae

(2) Chlorophyceae

(3) Rhodophyceae
(4) Cyanophyceae

27) Which of the following adaptation is not observed in *Pinus* needle leaf ?

(1) Thick cuticle
(2) Reduced surface area
(3) Sunken stomata
(4) Absence of stomata

28) "They occur at great depth in oceans where little light penetrates." Above character best suited to :

(1) Red algae
(2) Brown algae
(3) Green algae
(4) Gymnosperm

29) The prothallus of pteridophytes is:

(1) Conspicuous, large, multicellular
(2) Inconspicuous, small, multicellular, free-living
(3) Unicellular and dependent
(4) Parasitic

30) Megasporophylls are tightly packed in cones of:

(1) Female *Cycas*
(2) *Pteris* and *Adiantum*
(3) Female *Marchantia*
(4) None of these

31) Bryophyta have :

(1) Sporophyte for long duration.
(2) Dominant phase of sporophyte which is parasite.
(3) Dominant phase of gametophyte which produce spores.
(4) Small sporophyte phase which is generally dependent on gametophyte.

32) Which one of the following sentence is not true for Bryophytes?

(1) Amphibians of the plant kingdom.
(2) Non vascular embryophytes
(3) Zygotic meiosis
(4) Gametophyte develops from germinate spores.

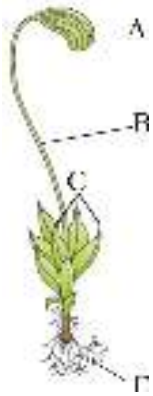
33) The main plant body of bryophytes is **___A___**. It produces **___B___**, hence is called a **___C___**. Fill in the blanks by choosing the correct option for A, B and C.

(1) A = diploid, B = spores, C = sporophyte
(2) A = haploid, B = gametes, C = gametophyte
(3) A = diploid, B = endospores, C = sporophyte
(4) A = haploid, B = conidia, C = gametophyte

34) "Gemmae" are multicellular green structures for vegetative propagation. These are found in :

(1) Prothallus of *Salvinia*
(2) Thallus of *Marchantia*
(3) Protonema of *Funaria*
(4) Prothallus of Fern

35) Identify the A, B, C and D respectively :



(1) Seta, Leaves, Capsule, Rhizoids
(2) Leaves, Capsule, Seta, Rhizoids
(3) Capsule, Seta, leaves, Rhizome
(4) Capsule, Seta, leaves, Rhizoids

36) Vegetative reproduction in mosses takes place by:-

(1) Fragmentation
(2) Budding in secondary protonema
(3) Spore formation
(4) Both (1) & (2)

37) Peat moss is :

(1) *Funaria*
(2) Fern
(3) Algae
(4) *Sphagnum*

38) Which structure is formed in the life cycle of moss, between spore and main plant ?

(1) Prothallus
(2) Embryo
(3) Spore mother cell
(4) Protonema

39) In moss reduction division takes place in :-

(1) Archegonia
(2) Capsule
(3) Antheridium
(4) At the tip of rhizoids

40)

Branched chain lipids occur in the cell membranes of :

(1) Archaebacteria
(2) Mycoplasma
(3) Actinomycetes
(4) Streptomyces

41)

Smallest known moneran lacking cell wall are :

(1) Spirochaete
(2) Mycoplasma
(3) Cyanobacteria
(4) Archaebacteria

42)

'Contagium vivum fluidum' (i.e., living fluid infectant) term has given by :

(1) Mayer
(2) Ivanowsky
(3) Beijerinck
(4) Bawden and Pirie

43)

Match the following and select the correct combination from the options given below.

	Column I (Kingdom)		**Column II (Class)**
a.	Plantae	1.	Archaebacteria
b.	Fungi	2.	Euglenoids
c.	Protista	3.	Phycomycetes

d.	Monera	4.	Algae

(1) a – 4, b – 3, c – 2, d – 1

(2) a – 1, b – 2, c – 3, d – 4

(3) a – 3, b – 4, c – 2, d – 1

(4) a – 2, b – 3, c – 4, d – 1

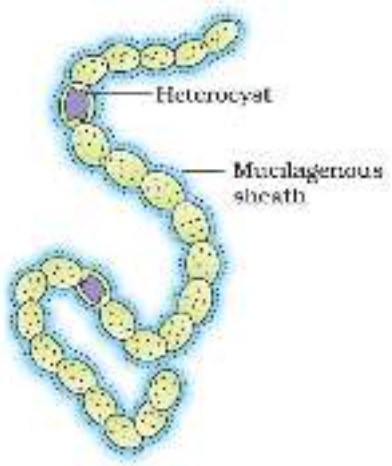
44) Lichens are :-

(1) Made up of algal component phycobiont and fungal component mycobiont.

(2) Found in polluted areas.

(3) Placed in kingdom-plantae according the R.H. Whittaker.

(4) Heterotrophic only.



45) Recognise the figure and find the incorrect option :

(1) Denitrification in specialised cell Heterocyst.

(2) This is filamentous BGA included in kingdom monera.

(3) It has 'chl a' similar to green plants.

(4) Colonies are surrounded by gelatinous sheath.

46) Select the correct combination of the statement (a–d) regarding the characteristics of certain organisms :
(a) Methanogens are Archaebacteria which produce methane in marshy areas.
(b) *Nostoc* is filamentous blue-green algae which fixes atmospheric nitrogen.
(c) Chemosynthetic autotropic bacteria synthesize cellulose from glucose.
(d) Mycoplasma lack a cell wall and can survive without oxygen.
The correct statements are :

(1) (a), (b), (c)

(2) (b), (c), (d)

(3) (a), (b), (d)

(4) (b), (c)

47) Infoldings of plasma membrane in bacteria are called :-

(1) Episomes

(2) Plasmid
(3) Pili
(4) Mesosomes

48) Which bacteria are utilized in Gober gas plant ?

(1) Methanogens
(2) Nitrifying bacteria
(3) Ammonifying bacteria
(4) Denitrifying bacteria

49) Which of following pair shows heterotrophic mode of nutrition ?

(1) *Chlorobium* and *Nitrosomonas*
(2) *Chlorella* and Cyanobacteria
(3) *Rhizobium* and *Lactobacillus*
(4) *Anabaena* and *Rhodospirillium*

50) Organism of which kingdom feed like animals (holozoic) and perform photosynthesis like plants ?

(1) Monera
(2) Protista
(3) Fungi
(4) Animalia

51) Armoured cell wall and biflagellated cells are characteristic of :-

(1) Chrysophyta
(2) Pyrrophyta
(3) Euglenophyta
(4) Cyanophyta

52) Dinoflagellates are called fire algae due to which character :-

(1) They appear like fire due to pigments
(2) They produce fire due to friction
(3) They occur on burnt places
(4) They show biolumineiscence

53) Read statements A to D :
(A) Golden Algae are classified under chrysophytes.
(B) *Gonyaulax* & *Diatoms* are classified under Dinoflagellates
(C) In Diatoms the cell wall form thin overlapping shell, which fit together as in soap box.
(D) Most of the Dinoflagellates have two flagella.
How many statement are wrong ?

(1) one

(2) two
(3) three
(4) four

54) **Statement-I :** Spore of slime moulds survives for years.
Statement-II : Spores of slime moulds posses true walls.

(1) Both statement are wrong.
(2) Only statement-I is wrong.
(3) Both statement are correct.
(4) Only statement-II is wrong.

55) Which of the following pigment is present in Euglenoid but lack in dinoflagellates and diatoms ?

(1) Chlorophyll 'a'
(2) Chlorophyll 'c'
(3) Chlorophyll 'b'
(4) Carotenes

56) Choose the incorrect statement about the given organism ? 

(1) Two, unequal flagella are found.
(2) It dies, when deprived of sunlight.
(3) Member of kingdom Protista.
(4) Photosynthetic pigments are similar to those of higher plants.

57) Phycomycetes produces :-

(1) Aplanospores, Zoospores, Zygospores
(2) Oospores, Zygospores, Ascospores
(3) Ascospores, Basidiospores, Oospores
(4) Ascospores, Basidiospores, Oospores

58) _____ is a parasite on mustard.

(1) *Albugo*
(2) *Puccinia*
(3) Yeast
(4) *Ustilago*

59) Find the incorrect statement about fungi -

(1) They show a great diversity in morphology and habitats.

(2) Fungi are cosmopolitan and occur in air, water, soil and as parasites too.

(3) Dikaryophase occur in Ascomycetes and Basidiomycetes.

(4) *Rhizopus* belongs to group basidiomycetes.

60) Identify the figures with features and select the correct option :-



(a)



(b)



(c)

(i) The mycelium is aseptate

(ii) Asexual spores are generally not found

(iii) Sexual spores are produced endogenously.

(1) a–i, b–ii, c–iii

(2) a–i, b–iii, c–ii

(3) a–ii, b–i, c–iii

(4) a–iii, b–ii, c–i

61) Match Column I with Column II.

Column I		**Column II**	
A	Porifera	i	Flame cells
B	Coelenterata	ii	Canal system
C	Platyhelminthes	iii	Radial symmetry
D	Annelida	iv	Metameric segmentation

(1) A-ii, B-iii, C-i, D-iv

(2) A-i, B-ii, C-iii, D-iv

(3) A-iii, B-ii, C-iv, D-i

(4) A-ii, B-iv, C-iii, D-i

62) Match the organisms with their phylum.

Organism		**Phylum**	
A	Sycon	i	Aschelminthes
B	Hydra	ii	Porifera
C	Ascaris	iii	Coelenterata
D	Nereis	iv	Annelida

(1) A-i, B-iii, C-ii, D-iv

(2) A-iii, B-ii, C-iv, D-i

(3) A-ii, B-iii, C-i, D-iv

(4) A-ii, B-iv, C-iii, D-i

63) Match the characteristic with the animal group.

Column-I		Column-II	
A	Cnidoblasts	i	Ctenophora
B	Comb plates	ii	Coelenterata
C	Pseudocoelom	iii	Aschelminthes
D	Setae	iv	Annelida

(1) A-i, B-ii, C-iv, D-iii

(2) A-ii, B-iv, C-iii, D-i

(3) A-iii, B-ii, C-i, D-iv

(4) A-ii, B-i, C-iii, D-iv

64) Match the feature with the phylum.

Column-I		Column-II	
A	Cellular level of organisation	i	Platyhelminthes
B	Tissue level of organisation	ii	Porifera
C	Acoelomate	iii	Annelida
D	True coelom	iv	Coelenterata

(1) A-iii, B-ii, C-iv, D-i

(2) A-ii, B-i, C-iii, D-iv

(3) A-ii, B-iv, C-i, D-iii

(4) A-i, B-ii, C-iii, D-iv

65) Consider the following statements :
A. All Poriferans are aquatic.
B. Most Poriferans are marine.
C. Canal system is present in Porifera.
D. Sponges possess organ level of organisation.
Find the correct statements -

(1) A, C and D only

(2) A, B and C only

(3) B and D only

(4) All are correct

66) Which statements are correct?
A. Hydra possesses cnidoblasts.
B. Corals belong to Coelenterata.
C. Ctenophores possess colloblasts.

D. Ctenophores possess cnidoblasts.

(1) A, B and C

(2) B, C and D

(3) A and D

(4) All are correct

67) Which statements are true about Platyhelminthes ?
A. Platyhelminthes are triploblastic.
B. They are bilaterally symmetrical.
C. They possess pseudocoelom.
D. They are dorsoventrally flattened.

(1) B and C

(2) A, C and D

(3) A, B and D

(4) All are correct

68) Which statements are correct?
A. Ascaris is dioecious.
B. Ascaris possesses pseudocoelom.
C. Earthworm has a closed circulatory system.
D. Earthworm is dioecious.

(1) A, C and D

(2) B, C and D

(3) A, B and C

(4) All are correct

69) Which statements are correct?
A. Segmentation is present in Annelida.
B. Closed blood vascular system is present in chordata.
C. True coelom is present in mallusca.
D. Flame cells are present in cephalochordata for respiration.

(1) A, B and C

(2) B, C and D

(3) A and D

(4) All are correct

70) **Assertion (A) :** Poriferans are multicellular animals.
Reason (R) : They possess a cellular level of organisation.

(1) Both A and R are true, and R correctly explains A.

(2) Both true, but R is not the correct explanation.

(3) A true, R false.

(4) A false, R true.

71) **Assertion (A) :** Platyhelminthes are called flatworms.
Reason (R) : Their body is dorsoventrally flattened.

(1) Both A and R are true, and R correctly explains A.
(2) Both A and R are true, but R is not the correct explanation.
(3) A true, R false.
(4) A false, R true.

72) **Assertion (A) :** Ctenophores show bioluminescence.
Reason (R) : They possess eight rows of ciliary comb plates.

(1) Both A and R are true, and R correctly explains A.
(2) Both A and R are true, but R is not the correct explanation.
(3) A true, R false.
(4) A false, R true.

73) Which phylum possesses choanocytes?

(1) Platyhelminthes
(2) Coelenterata
(3) Annelida
(4) Porifera

74) Which phylum is known as roundworms?

(1) Platyhelminthes
(2) Aschelminthes
(3) Annelida
(4) Ctenophora

75) Which of the following exhibits metameric segmentation and possess nephridia for excretion ?

(1) Hydra
(2) Ascaris
(3) Earthworm
(4) Taenia

76) Match Column I with Column II.

Column-I		**Column-II**	
A	Arthropoda	i	Jointed appendages
B	Mollusca	ii	Muscular foot
C	Echinodermata	iii	Water vascular system
D	Hemichordata	iv	Proboscis

(1) A-i, B-iii, C-ii, D-iv

(2) A-ii, B-i, C-iv, D-iii

(3) A-i, B-ii, C-iii, D-iv

(4) A-iv, B-ii, C-i, D-iii

77) Match the organism with its phylum.

Column-I		**Column-II**	
A	Pila	i	Arthropoda
B	Starfish	ii	Mollusca
C	Cockroach	iii	Echinodermata
D	Balanoglossus	iv	Hemichordata

(1) A-iii, B-ii, C-iv, D-i

(2) A-ii, B-i, C-iii, D-iv

(3) A-i, B-iii, C-ii, D-iv

(4) A-ii, B-iii, C-i, D-iv

78) Match the vertebrate class with its characteristic.

Column-I		**Column-II**	
A	Chondrichthyes	i	Swim bladder present
B	Osteichthyes	ii	Cartilaginous skeleton
C	Amphibia	iii	Moist glandular skin
D	Aves	iv	Pneumatic bones

(1) A-ii, B-i, C-iii, D-iv

(2) A-i, B-ii, C-iii, D-iv

(3) A-ii, B-iii, C-iv, D-i

(4) A-iv, B-ii, C-i, D-iii

79) Match the group with the example.

Column I		**Column II**	
A	Urochordata	i	Branchiostoma
B	Cephalochordata	ii	Herdmania
C	Cyclostomata	iii	Petromyzon
D	Mammalia	iv	Whale

(1) A-i, B-ii, C-iv, D-iii

(2) A-ii, B-i, C-iii, D-iv

(3) A-ii, B-iii, C-i, D-iv

(4) A-iii, B-i, C-ii, D-iv

80) Consider the following statements :
A. Arthropods possess jointed appendages.
B. Arthropods have an open circulatory system.
C. Cockroach belongs to Arthropoda.
D. Arthropods possess gills, book gills, book lungs for respiration.

(1) B, C and D only
(2) A and D only
(3) A, B and C only
(4) All are correct

81) Which of the following statements are correct?
A. Echinoderms are exclusively marine.
B. Adults echinoderms show radial symmetry.
C. Larvae of Echinodermata are bilaterally symmetrical.
D. Echinodermata possess a muscular foot.

(1) A, B and C
(2) B and D
(3) A and D
(4) All of these

82) Which statements are correct for Chordates?
A. Notochord present at least in embryonic stage.
B. Dorsal hollow nerve cord present.
C. Pharyngeal gill slits present during some stage.
D. Ventral solid nerve cord present.

(1) B, C and D
(2) A, B and C
(3) A and D
(4) All of these

83) Which statements are correct?
A. Chondrichthyes possess placoid scales.
B. Osteichthyes possess bony endoskeleton.
C. Swim bladder is absent in Chondrichthyes.
D. Pisces have four-chambered hearts.

(1) B, C and D
(2) A, C and D
(3) A, B and C
(4) All of these

84) Which statements are correct?
A. Birds are warm-blooded.
B. Mammals possess mammary glands.
C. Amphibians possess feathers.

D. Reptiles have dry cornified skin.

(1) A, B and C
(2) B, C and D
(3) A, B and D
(4) All of these

85) **Assertion (A) :** Mammals possess mammary glands.
Reason (R) : Mammary glands secrete milk for nourishing the young ones.

(1) Both A and R are true, and R is the correct explanation of A.
(2) Both are true, but R is not the correct explanation.
(3) A is true but R is false.
(4) A is false but R is true.

86) **Assertion (A) :** Amphibians are called "double-life animals".
Reason (R) : They spend part of their life in water and part on land.

(1) Both A and R are true, and R is the correct explanation of A.
(2) Both are true, but R is not the correct explanation.
(3) A is true but R is false.
(4) A is false but R is true.

87) **Assertion (A) :** Birds possess pneumatic bones.
Reason (R) : Pneumatic bones reduce body weight and aid in flight.

(1) Both A and R are true, and R is the correct explanation of A.
(2) Both are true, but R is not the correct explanation.
(3) A is true but R is false.
(4) A is false but R is true.

88) Which animal belongs to Cyclostomata?

(1) Scoliodon
(2) Petromyzon
(3) Labeo
(4) Frog

89) Which class possesses cycloid and ctenoid scales ?

(1) Osteichthyes
(2) Chondrichthyes
(3) Amphibia
(4) Reptilia

90) Which of the following is a mammal?

(1) Whale
(2) Penguin
(3) Crocodile
(4) Rohu

ANSWER KEYS

PHYSICS

Q.	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20
A.	3	1	2	1	3	2	4	2	4	4	1	1	2	2	3	4	2	2	4	2
Q.	21	22	23	24	25	26	27	28	29	30	31	32	33	34	35	36	37	38	39	40
A.	1	3	4	1	3	2	3	1	4	2	2	4	1	1	1	2	1	3	3	3
Q.	41	42	43	44	45															
A.	2	3	4	4	3															

CHEMISTRY

Q.	46	47	48	49	50	51	52	53	54	55	56	57	58	59	60	61	62	63	64	65
A.	2	3	1	2	1	2	3	4	1	1	1	2	1	1	3	2	2	4	2	3
Q.	66	67	68	69	70	71	72	73	74	75	76	77	78	79	80	81	82	83	84	85
A.	1	4	4	3	1	3	4	3	4	2	3	2	1	3	4	3	2	2	3	3
Q.	86	87	88	89	90															
A.	4	1	3	4	2															

BIOLOGY

Q.	91	92	93	94	95	96	97	98	99	100	101	102	103	104	105	106	107	108	109	110
A.	3	1	2	4	1	3	2	3	4	2	2	1	4	1	1	1	4	4	2	2
Q.	111	112	113	114	115	116	117	118	119	120	121	122	123	124	125	126	127	128	129	130
A.	3	4	3	4	2	3	4	1	2	4	4	3	2	2	4	4	4	4	2	1
Q.	131	132	133	134	135	136	137	138	139	140	141	142	143	144	145	146	147	148	149	150
A.	2	3	1	1	1	3	4	1	3	2	2	4	1	3	3	2	1	1	4	2
Q.	151	152	153	154	155	156	157	158	159	160	161	162	163	164	165	166	167	168	169	170
A.	1	3	4	3	2	1	3	3	1	2	1	2	4	2	3	3	4	1	2	4
Q.	171	172	173	174	175	176	177	178	179	180										
A.	1	2	3	3	1	1	1	2	1	1										

SOLUTIONS

PHYSICS

1)

Explaining : We are given balloon and man is at rest and in equilibrium. If the man starts climbing and reaches at the top of rope, Choose the correct statement(s)
Concept: This question is based on Motion of COM.
Solution:



$(\Delta y)_{centre\ of\ mass} = 0$

$$\Delta y_{cm} = \frac{m_1\Delta y_1 + m_2\Delta y_2}{m_1 + m_2}$$

$$0 = \frac{m(L-y) + M(-y)}{M+m}$$

$mL - my = My$

$$mL = y(M+m) \Rightarrow \frac{m}{M} = \frac{y}{L-y}$$

$$y = \frac{mL}{M+m}$$

y = descend of balloon

ascend of man = L – y

$$= L - \frac{mL}{M+m} = \frac{ML}{M+m}$$

 man ascends by L – y

 his potential energy increases by mg(L – y)

$$= \Delta(PE)_{man} = mg\,[L-y]$$

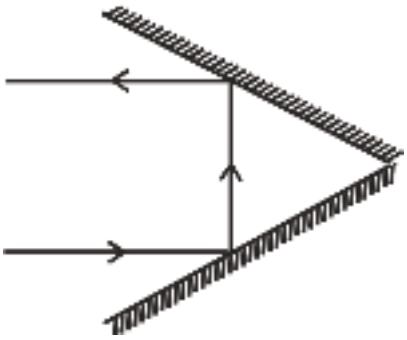
$$= \frac{mg[ML]}{M+m}$$

and $(\Delta PE)_{balloon} = Mg(-y) = -\frac{mML}{M+m}g$

$\Delta PE_{Total} = 0$

Final Answer: The correct option is (3)

2) **Explaining :** To find magnitude of net impulse provided by both walls.
Concept : This question is based on concept of impulse.
Solution :



Impulse = 2 ×10 – (–2 × 10) = 40 Ns

Final Answer : (1)

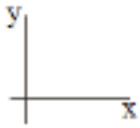
3) **Explaining :** To find the angle ϕ.

Concept : This question is based on concept of coefficient of restitution.

Solution :

In y direction

v sin ϕ = e u sin45º ....(i)



In x direction

v cos ϕ = u cos 45° ....(ii)

eq. (i) ÷ eq. (ii)

$\tan \phi = \frac{1}{\sqrt{3}}$; ϕ = 30º

Final Answer : (2)

4) **Explaining :** To find the coefficient of restitution of impact.

Concept : This question is based on concept of coefficient of restitution.

$$V_{Avg.} = \frac{2L}{\frac{L}{V} + \frac{L}{ev}} = \frac{2V}{3}$$

Solution :

$V_{avg.} = \frac{2L}{\frac{eL+L}{ev}} = \frac{2V}{3}$

$\frac{2L\,(eV)}{(e+L)L} = \frac{2V}{3}$

3e = e + 1

2e = 1

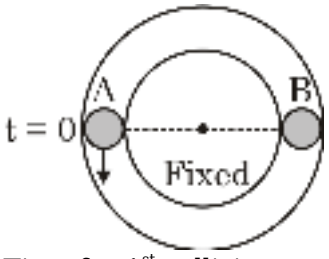
$e = \frac{1}{2} = 0.5$

Final Answer : (1)

5) **Explaining :** To find the time at which second collision occurs.

Concept : This question is based on concept of collision.

Solution :



Time for 1st collision

$t_1 = \frac{\pi R}{v} = \frac{\pi(S)}{10} = \frac{\pi}{2}$ sec

For 1st collision :- $u_1 = 10$ m/s $u_2 = 0$

$m_1 = m \quad m_2 = 2m$ & $e = \frac{1}{2}$

using momentum conservation :-

$10m = mv_1 + 2mv_2$

$\Rightarrow v_1 + 2v_2 = 10 \quad \rightarrow \quad (1)$

$e = \frac{v_2 - v_1}{u_1 - u_2} \Rightarrow \frac{1}{2} = \frac{v_2 - v_1}{10 - 0}$

$\Rightarrow v_2 - v_1 = 5 \quad \rightarrow \quad (2)$

From eqn (1) & eqn (2)

$v_2 = 5$; $v_1 = 0$

time b/w Ist & 2nd collision

$= \frac{2\pi R}{v_{Rel}} = \frac{2\pi \times 5}{5} = 2\pi$

So the time at which 2nd collision occur

$= 2\pi + \frac{\pi}{2} = \frac{5\pi}{2}$ sec

Final Answer : (3)

6) **Explain Question:**

The question asks to find the ratio $\left(\frac{m_1}{m_2}\right)$ when two balls collision elastically.

Concept:

Head on elastic collision (COLM).

Solution:

As $v_1 = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)u_1 + \frac{2m_1u_2}{m_1 + m_2}$

As $u_2 = 0$ and $v_1 = \frac{v}{3}$

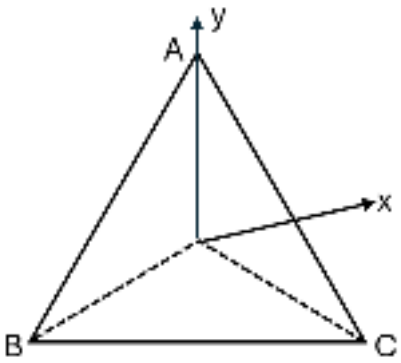
$\Rightarrow \frac{v}{3} = \left(\frac{m_1 - m_2}{m_1 + m_2}\right)v$

Solving we get $\frac{m_1}{m_2} = 2$

Final Answer: Option (2)

7) **Explanation:** A triangular wire frame is given. In different options different sides are remove then position of COM will be -

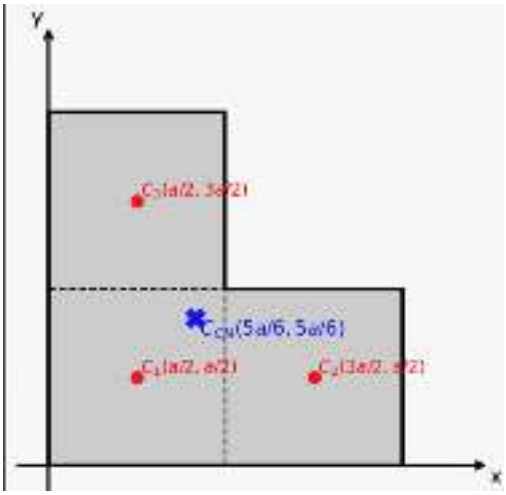
Concept: Calculation of COM.
Solution:



If AB is removed COM will be in 4th quadrant.
If BC is removed COM will be on y-axis.
If AC is removed COM will be in 3rd quadrant.
Note: COM will be towards the heavy mass.
Final Ans: Option (4)

8)

Question Explanation:
The question asks to find the coordinates of the center of mass of an L-shaped uniform lamina.
Concept:
Center of mass of a system of particles/composite bodies.
Solution:



The uniform L-shaped lamina can be split into three identical squares of side length {a}. Since the mass per unit area is uniform, each square has the same mass {m}. The total center of mass is the weighted average of the centers of mass of these three individual squares.
Calculation:
Divide the shape into three squares, each of area $A=a^2$ and mass m :
Square 1 (bottom-left): Center at $(x_1, y_1) = \left(\frac{a}{2}, \frac{a}{2}\right)$
Square 2 (bottom-right): Center at $(x_2, y_2) = \left(\frac{3a}{2}, \frac{a}{2}\right)$
Square 3 (top-left): Center at $(x_3, y_3) = \left(\frac{a}{2}, \frac{3a}{2}\right)$
Calculate the X -coordinate of the center of mass (X_{CM}) :

$$X_{CM} = \frac{mx_1 + mx_2 + mx_3}{3m} = \frac{\frac{a}{2} + \frac{3a}{2} + \frac{a}{2}}{3} = \frac{\frac{5a}{2}}{3} = \frac{5a}{6}$$

Calculate the Y -coordinate of the center of mass (Y_{CM}) :

$$Y_{CM} = \frac{my_1 + my_2 + my_3}{3m} = \frac{\frac{a}{2} + \frac{a}{2} + \frac{3a}{2}}{3} = \frac{\frac{5a}{2}}{3} = \frac{5a}{6}$$

Matching the calculated coordinates with the given options shows that both Option 1 and Option 2 are correct statements.

Final Answer: (2)

9) **Explaining :** To find on which line COM doesn't lie on.
Concept : This question is based on centre of mass.
Solution :

$$X_{CM} = \frac{2 \times 15}{5} = 6$$

$$Y_{CM} = \frac{3 \times 20}{5} = 12$$

Final Answer : (4)

10)

Explanation Question : This problem asks to find the x-coordinate of the center of mass for an infinite series of rods placed as shown in figure.
Concept : This question is based on calculation of COM.
Solution : Since each rod has uniform mass density, the center of mass for a rod of length L placed with in left end as x = 0 is as $x = \frac{L}{2}$ from left

$$x_{COM} = \frac{m_1x_1 + m_2x_2 + ---}{m_1 + m_2 + ----}$$

$$= \frac{(PL)\left(\frac{L}{2}\right) + \left(P\frac{L}{2}\right)\left(\frac{L}{4}\right) + \left(P\frac{L}{4}\right)\left(\frac{L}{8}\right) + -----}{(PL) + \left(P\frac{L}{2}\right) + \left(P\frac{L}{4}\right) + ----}$$

$$= \frac{L\left[\frac{1}{2} + \frac{1}{8} + \frac{1}{32} + ----\right]}{\left[1 + \frac{1}{2} + \frac{1}{4} + ----\right]}$$

$$= \frac{L\left[\frac{1/2}{1-1/4}\right]}{\left[\frac{1}{1-1/2}\right]}$$

$x_{COM} = \frac{L}{3}$ left of (L,0)

 coordinate of COM from origin $x_{COM} = \frac{2L}{3} = \frac{2 \times 6}{3} = 4$ units

Final Answer : (4)

11) **Explain question :** Half ring and half disc arranged as shown. Find the Y-coordinate of the system.
Concept : This question is based on calculation of center of mass.
Solution : The system is symmetrical about the Y axis.

$$y_{cm} = \frac{m \times \frac{2R}{\pi} + m\left(-\frac{4R}{3\pi}\right)}{m + m}$$

$$\boxed{y_{cm} = \frac{R}{3\pi}}$$

Final answer : 1

12) **Explaining :** To choose the **CORRECT** statement.
Concept : This question is based on moment of inertia.
Solution : Moment of inertia is maximum if masses are farther from the axis and vice-versa
Final Answer : (1)

13) **Explaining :** To choose the correct free body diagram of the block.
Concept : This question is based on free body diagram.
Solution : Normal shfit towards point of rotation.
Final Answer : (2)

14) **Explain the question :** In this question, a wheel-an-axle system is shown, and we are asked to determine the condition for the system to remain in static equilibrium.
Concept: This question is based on the conditions of static equilibrium.
Solution: Using torques balance about center of the wheel.
$am_1g = bm_2g$
$am_1 = bm_2$
Final Ans: 2

15) **Explaining :** To find the moment of inertia of the system (in kg-m^2) of seven pennies about an axis that passes through the center of the central penny and is normal to the plane of the pennies.
Concept : This question is based on concept of parallel axis theorem.
Solution : $I_{sys} = \frac{mr^2}{2} + \left[\frac{mr^2}{2} + m(2r)^2\right] \times 6$

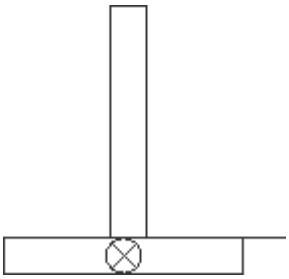
$= \frac{55}{2} mr^2 = 55$

Final Answer : (3)

16) **Explaining :** To find the value of the unknown mass m (in Kg).
Concept : This question is based on concept of meter scale the mass of the given object by the principle of moments.
Solution :
$T_{net} = 0$
$16 \times 6 = m \times 8$
$m = 12$ kg

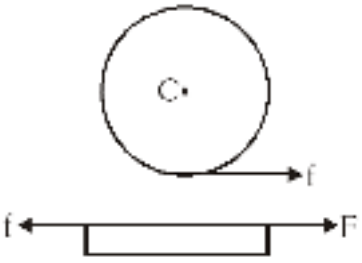
Final Answer : (4)

17) **Explaining :** To find the moment of inertia of rods about an axis passing through O and perpendicular to the plane of rods.
Concept : This question is based on concept of moment of inertia.
Solution :



$$I = \frac{ML^2}{3} + \frac{ML^2}{12}$$
$$I = \frac{4ML^2 + ML^2}{12}$$
$$I = \frac{5ML^2}{12}$$
Final Answer : (2)

18) **Explaining :** To find if the plank is pulled with a constant force F. The friction force can be given by which of the following diagrams.
Concept : This question is based on concept of rolling motion.
Solution :

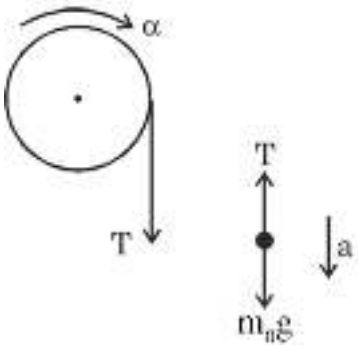
Friction force opposes relative slipping.  **Final Answer : (2)**

19) **Explanation:**
In this question, a uniform rod of mass m and length is suspended horizontally by two ideal strings attached at its ends as shown in the figure. The tensions in the two strings are T_1 and T_2. The relation between these tensions and the weight of the rod is to be identified.
Concept:
This question is based on conditions of equilibrium of a rigid body, involving balance of forces and balance of torques.
Solution :
$T_1 + T_2 = mg$ (Force balance)
Torque about one end of rod
$$T_2 \ell \cos 30^\circ - mg\frac{\ell}{2}\cos 30^\circ = 0$$
So $$T_1 = T_2 = \frac{mg}{2}$$

Final answer : Option(4)

20)

Explanation Question : We are given uniform solid cylinder and we have to find acceleration of particle.
Concept: This question is based on torque analysis.
Solution:



$m_0g - T = m_0a$

$$TR = \frac{mR^2}{2}\alpha$$

$a = R\alpha$

$$T = \frac{ma}{2}$$

$$m_0g = \left(\frac{m}{2} + m_0\right)a$$

$$a = \frac{2m_0g}{m + 2m_0}$$

Final Answer : The correct option is (2)

21)

Explanation Question: The problem asks to find the sum of moment of inertia, I_1,I_2, I_3 for a disc of mass m and radius R.

Concept: Moment of inertia
Solution:
We know that

$$I_3 = \frac{mR^2}{2}$$

According to perpendicular axis theorem $I_3 = I_1 + I_2$

$I_1 + I_2 + I_3 = 2I_3 = mR^2$
Final Answer option : (1)

22)

Question Explanation : find the linear acceleration of a solid sphere rolling down a 30◦ inclined plane without slipping.

Concept: Rolling Motion on an Inclined Plane

Formula :

$$a = \frac{g\sin\theta}{1+\frac{K^2}{R^2}}$$

Solution:

$$a = \frac{g\sin\theta}{1+\frac{K^2}{R^2}}$$

$$\Rightarrow a = \frac{g\sin 30^\circ}{1+\frac{2}{5}}$$

$$\Rightarrow a = \frac{10\times\frac{1}{2}}{\frac{7}{5}}$$

$$a = \frac{25}{7}\ ms^{-2}$$

Final Answer: Option (3)

23)

Question Explanation:
The question asks to find the linear velocity of mass B.

Concept:
Conservation of Mechanical Energy and Rotational Kinematics.

Solution:
Moment of Inertia (I):

$$I = m\left(\frac{\ell}{3}\right)^2 + m\left(\frac{2\ell}{3}\right)^2 + m\ell^2 = \frac{14}{9}m\ell^2$$

Loss in Potential Energy (ΔU):

$$\Delta U = mg\left(\frac{\ell}{3}\right) + mg\left(\frac{2\ell}{3}\right) + mg\ell = 2mg\ell$$

Angular Velocity (ω):

$$\frac{1}{2}I\omega^2 = \Delta U \Longrightarrow \frac{1}{2}\left(\frac{14}{9}m\ell^2\right)\omega^2 = 2mg\ell \Longrightarrow \omega = \sqrt{\frac{18g}{7\ell}}$$

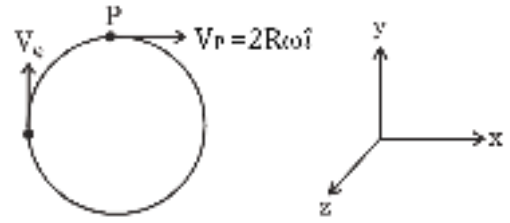
Velocity of $B\,(v_B)$:

$$v_B = \left(\frac{2\ell}{3}\right)\omega = \sqrt{\frac{4\ell^2}{9}\cdot\frac{18g}{7\ell}} = \sqrt{\frac{8g\ell}{7}}$$

Final Answer: Option (4)

24)

Explain Question : A ball is pure rolling on ground and we have to find ratio.
Concept : This question is based on combined rotation and translation motion.
Solution :



$V_Q = R\omega\hat{j} + R\omega\hat{i}$

$\therefore \frac{|V_P|}{|V_Q|} = \frac{\sqrt{2}}{1}$

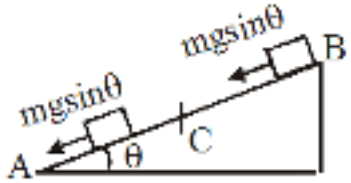
Final answer : (1)

25)

Question Explanation : In the experiment the beam balance is in the state of equilibrium in two ways as shown. We need to find the unknown mass m.
Concept : Equilibrium of rigid bodies.
Solution :
mg ℓ_2 = 16 gℓ_1 .....(i)
mg ℓ_1 = 4gℓ_2 .....(ii)
$\frac{16}{m} = \frac{m}{4}$ ⇒ m = 8 kg
Final answer : Option(3)

26) **Explaining :** To find the value of x.
Concept : This question is based on concept of vertical circular motion.
Solution :
$v = \sqrt{u^2 - 2gL}$
$\Delta v = \sqrt{u^2 + v^2}$
$\Delta v = \sqrt{u^2 + v^2 - 2gL}$
$\Delta v = \sqrt{2u^2 - 2gL}$
$\Delta v = \sqrt{2(u^2 - gL)}$ x = 2
Final Answer : (2)

27) **Explaining :** To find the coefficient of friction.
Concept : This question is based on concept of motion on inclined plane.
Solution :



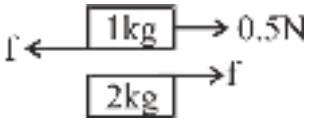
Apply work energy theorem
mgsinθ (AC + 2AC) – μmg cosθAC = 0
μ = 3tanθ
Final Answer : (3)

28) **Explaining :** To find the work done on the body till anytime t.

Concept : This question is based on concept of work energy theorem.
Solution :
W = ΔKE
All
$= \frac{1}{2}mv^2 = \frac{1}{2}m(at)^2$
$= \frac{1}{2}\frac{mv_0^2}{t_0^2}t^2$
Final Answer : (1)

29) **Question Explanation:** The question asks to find the value of N.
Concept: Conservation of Mechanical Energy.
Solution:
Potential energy U = $\frac{x^4}{4} - \frac{x^2}{2}$
For maximum kinetic energy, potential energy of a particle should be minimum.
For minimum value of U, $\frac{dU}{dx} = 0$ and $\frac{d^2U}{dx^2} > 0$
For U_{min} : $\frac{dU}{dx} = x^3 - x = 0 \Rightarrow x = \pm 1.$
After calculation at x = ±1
$U_{min} = -\frac{1}{4}J$
$TE = KE_{max} + PE_{min}$
$\Rightarrow KE_{max} = TE - PE_{min}$
$\Rightarrow \frac{1}{2}mv_{max}^2 = 2 - \left(-\frac{1}{4}\right)$ $v_{max} = \frac{3}{\sqrt{2}}m/s$
Final Answer: (4)

30) **Explaining :** To find the work done by the lower block on the upper block for a displacement of 3 m of the upper block.
Concept : This question is based on concept of two block system.
Solution :



0.5 = 3a ⇒ a = 5/30
f = 2a = 2(5/30)= 1/3N
f_{max} = 0.1 × 10 = 10N
 f = 1/3 N will act
work done by lower block
= –f(s)
= –1/3(3) = –1 Joule
Final Answer : (2)

31)

Explanation Question: We are given acceleration system and we have to find work done by

force of gravity.
Concept: This question is based on calculation of work
Solution:

$a = \frac{40}{8} = 5\ m/s^2$

$s = \frac{1}{2}at^2 = 10m$

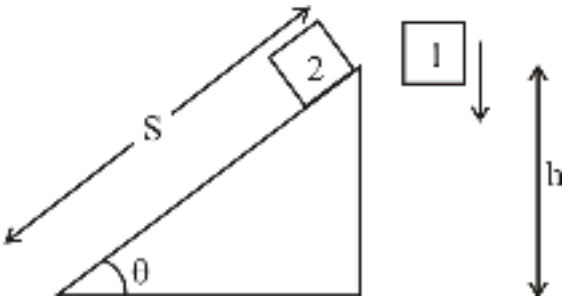
$\therefore W_g = 40 \times 10 = 400\ J$

Final Answer

The correct option is (2)

32)

Question Explanation : To analyze the motion for the block sliding down on a smooth inclined plane and another block which is released from the top.
Concept : This question is based on motion under gravity/free fall.
Solution :



Acceleration for the block sliding down a' = g sin θ
Acceleration for the block falling down a = g
length of the path travelled by block 2 = h/sin θ
length of the path travelled by block 1 = h
by conservation of energy

$\Rightarrow \frac{1}{2}mv^2 = mgh$ (potential energy depend only on height)

$v = \sqrt{2gh}$. Both will reach the ground with same speed.

But as distance is more for inclined path and acceleration (g sinθ) is less than g, hence block 2 will take more time than freely falling block.
hence speed are same for both but sliding block will take more time than free falling block to reach the ground.

Final Answer : Option (4)

33) **Explaining :** To find the decrease in gravitational potential energy of the system (in Joule).
Concept : This question is based on concept of gravitational potential energy.
Solution :

$U_i = m_1 g\ R \cos\frac{\alpha}{2} + m_2 g\ R \cos\frac{\alpha}{2}$

$U_f = m_1 gR + m_2 g\ R \cos\alpha$

$\therefore U_i - U_f$

$U_i = (2 \times 10 \times 0.5 \times 0.9) + (6 \times 10 \times 0.5 \times 0.9) = 36\ J$

$U_f = (2 \times 10 \times 0.5) + (6 \times 10 \times 0.5 \times 0.6) = 28\ J$

$\Delta U_i - U_f = (36 - 28) = 8$ J
Final Answer : (1)

34)

Question Explanation: We need to find the relation between W_1, W_2, W_3 for three different paths from A to B in a gravitational field.
Concept: Work done by gravitational force is conservative (path—independent).
Solution:
In a field produced only by mass M, the gravitational force is conservative. Hence the work done by it depends only on the initial and final positions of the particle, not on the path followed between them. Since in all three cases the particle goes from A to B, the work done along each path is the same.
Calculation: $W_1 = W_2 = W_3 = W_{A \to B}$
Final Answer: (1)

35) **Question Explanation :** A ball of mass m is fired horizontally from a spring-loaded gun (constant k) on a table of height h. It lands at a horizontal distance d. Find the initial compression x of the spring.
Concept: Conservation of Energy & Projectile Motion.
Solution: Time of flight :

$$t = \sqrt{\frac{2h}{g}}$$

Horizontal velocity (v) required to cover distance d in time t :

$$v = \frac{d}{t} = d\sqrt{\frac{g}{2h}}$$

Apply Conservation of Energy The potential energy stored in the spring converts to the kinetic energy of the ball.

$$\frac{1}{2}kx^2 = \frac{1}{2}mv^2$$

$$x^2 = \frac{m}{k}v^2$$

$$x = v\sqrt{\frac{m}{k}}$$

$$x = \left(d\sqrt{\frac{g}{2h}}\right)\sqrt{\frac{m}{k}}$$

$$x = d\sqrt{\frac{mg}{2hk}}$$

Final Answer: Option (1)

36)

Question Explanation : Find x from power of tension on block A.
Concept: This question is based on power.
Solution:
$2mg - T = 2ma$
$T - mg = ma$
$mg = 3ma$

$a = \frac{g}{3}$

$T = mg + \frac{mg}{3}$

$T = \frac{4mg}{3}$

$v = \frac{g \times 2}{3}$

$P = \frac{4mg}{3} \times \frac{2g}{3}$

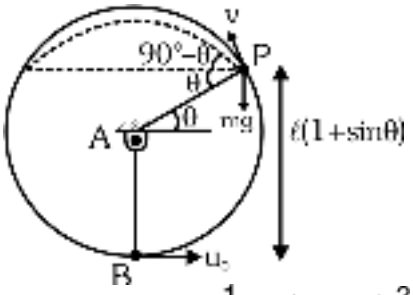
$P = \frac{8mg^2}{9}$

Final Answer : (2)

37) **Explain question :** This problem requires matching the nature of the pendulum (massless rod or string) and the initial velocity (u_0) to the maximum height attained by the bob.

Concept : Pendulum motion, Energy conservation.

Solution :



For (A) : $mgh = \frac{1}{2} m \left(\sqrt{3g\ell}\right)^2 \Rightarrow h = \frac{3\ell}{2}$

For (B) : $mgh = \frac{1}{2} m \left(\sqrt{4g\ell}\right)^2 \Rightarrow h = 2\ell$

For (C) : In this case bob just complete the circle so $h = 2\ell$

For (D) : As $\sqrt{2\ell g} < u_0 < \sqrt{5\ell g}$, the string will slack in upper half of the circle.

Assuming that string slacks when it makes an angle θ with horizontal.

At P, $T = 0$ so $mg\sin\theta = \frac{mv^2}{\ell}$

from energy conservation $\frac{1}{2} mu^{02} = \frac{1}{2} mv^2 + mg\ell (1+ \sin\theta) \Rightarrow \sin\theta = \frac{1}{3}$ & $v^2 = \frac{g\ell}{3}$

Maximum height attained from starting point = $\ell(1 + \sin\theta) + \frac{v^2\sin^2(90^\circ - \theta)}{2g} = \frac{4\ell}{3} + \frac{4\ell}{27} = \frac{40}{27}\ell$

Final answer : Correct option (1).

38)

QUESTION EXPLANATION

Assertion states: If work done by the force depends on path then force is said to be conservative.

Reason states: All the conservative forces are central forces. We must determine the truth of both statements and whether the reason explains the assertion.

CONCEPT

This question is based on Conservative and Non-conservative Forces

SOLUTION

Assertion is incorrect because a conservative force is path-independent , not path-dependent .For a conservative force, work done between two points is independent of path. It depends only on initial and final position ($W_c = 0$ in closed loop).
Reason is also incorrect because though many conservative forces (like gravitational, electrostatic) are central, not all central forces are conservative in every case.
FINAL ANSWER (Option 3)

39) **Explain the question:**
A particle moves in a circle with given angular velocity and angular acceleration; find total revolutions in 3 seconds.
Concept:
Angular motion with reversal of direction
Solution:
$\omega = w_0 + \alpha t$
$0 = 4 - 2t$
$t = 2$ s
So, the particle moves forward for 2 s, then reverses direction for the remaining 1 s.
Angular displacement in first 2 s:
$\theta_1 = \omega_0 t + (1/2)\alpha t^2$
$\theta_1 = 4\times 2 + (1/2)(-2)(2)^2$
$\theta_1 = 8 - 4 = 4$ rad
Angular velocity at reversal = 0
Angular displacement in next 1 s (starting from rest):
$\theta_2 = (1/2)\alpha t^2$
$\theta_2 = (1/2)(-2)(1)^2 = -1$ rad
Total angular distance travelled (magnitude):
$\theta_{total} = 4 + 1 = 5$ rad
Final Answer: (3)

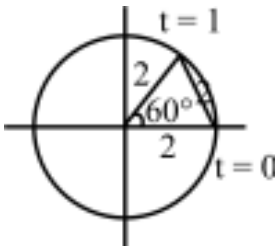
40)

Explain question : Angular displacement as a function of time is given we have to find average angular velocity and average velocity in given time.
Concept : This question is based on kinematics of circular motion
Solution :-

Average angular velocity = $\frac{\text{total angular displacement}}{\text{total time}}$

$= \frac{\left(\frac{\pi}{6} + \frac{\pi}{6}\right) - 0}{1}$

$< \omega > = \frac{\pi}{3}$ rad/sec.



average velocity = $\frac{\text{total displacement}}{\text{total time}}$

$< \vec{v} > = \frac{2}{1} = 2\text{m/sec}$

Final Answer : (3)

41) **Question Explanation:**
To find the value of x, we need to analyze the force acting on a vehicle in a banked curve.

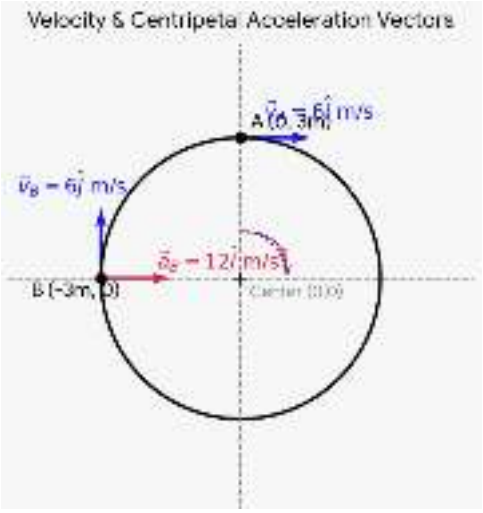
Concept :
This question is based on banking of roads.

Solution :
$V = \sqrt{Rg \tan \theta}$
$V = \sqrt{200 \times 10 \times 0.2}$
V = 20
In second condition
$V^2 = \left(\frac{\mu + \tan\theta}{1 - \mu \tan\theta}\right) Rg$
$(40)^2 = \left(\frac{\mu + 0.2}{1 - 0.2\mu}\right)(200 \times 10)$
$40 \times 40 = \frac{(\mu + 0.2)}{(1 - 0.24)} \times 2000$
⇒ 4 – 0.8 μ = 5μ + 1
4 – 1 = 5μ + 0.8μ
3 = 5.8 μ
$\mu = \frac{30}{58} = \frac{15}{29}$
Final answer :- (2)

42)

Question Explanation: We need to find the velocity vector and acceleration vector of the car when it reaches point B.

Concept: Kinematics of uniform circular motion and vector representation.
Solution:



From coordinates of point A(0,3 m), the radius of the circular track is:

$R = 3$ m

The constant speed of the car given by the velocity at A is:

$v = 6$ m/s

Since velocity at A points along $+\hat{\imath}$, the car rotates in a clockwise direction.
At point $\mathrm{B}(-3\ \mathrm{m}, 0)$, the tangent to the path points vertically upward along the positive y-axis:

$\vec{v}_B = v\hat{\jmath} = 6\hat{\jmath}$ m/s

Since speed is constant, the acceleration is purely centripetal and points directly toward the center of the circle (0,0) :

$\vec{a}_B = a_c(+\hat{\imath})$

The magnitude of the centripetal acceleration is:

$$a_c = \frac{v^2}{R}$$

Substituting the values:

$$a_c = \frac{6^2}{3} = \frac{36}{3} = 12\ \mathrm{m/s^2}$$

Expressing acceleration as a vector:

$\vec{a}_B = 12\hat{\imath}\ \mathrm{m/s^2}$

Final Answer: Option (3)

43)

Question Explanation : A particle starts from rest and moves in a circular path with a constant angular acceleration. We have to find the time when the magnitudes of its centripetal and tangential accelerations are equal.

Concept : This question is based on acceleration in non-uniform circular motion.

Solution :

Tangential acceleration, $a_T = R\alpha$ [R = radius of circular path]

α = angular acceleration = 16 rad./sec^2

Centripetal acceleration, $a_C = \frac{v^2}{R}$

By the equation, $v = u + at$

$v = a_T t$ $(u = 0)$

$\Rightarrow v = \mathbf{R}\alpha t$ [t = time]

$\Rightarrow a_c = \frac{v^2}{R} = \frac{\alpha^2 R^2 t^2}{R} = \alpha^2 R t^2$

$\Rightarrow$ the time at which both acceleration are equal

$\Rightarrow a_t = a_c \Rightarrow \alpha R = \alpha^2 R t^2$

$\Rightarrow t^2 = \frac{1}{\alpha} \Rightarrow t = \frac{1}{4}$ $(\because \alpha = 16$ rad./sec$^2)$

Final Answer : Option (4)

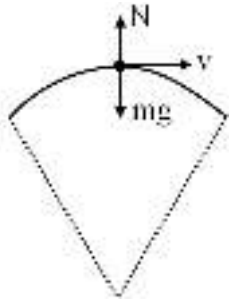
44) **Explain question :** The problem is about the normal contact force between motorcycle and bridge (in the form of circular arc) at the heighest form.

Concept : This question is based on dynamics of circular motion.

Solution :

Required centripital force equation is given by

$mg - N = \frac{mv^2}{r}$



$\Rightarrow$ Normal contact force (N)

$= mg - \frac{mv^2}{r}$

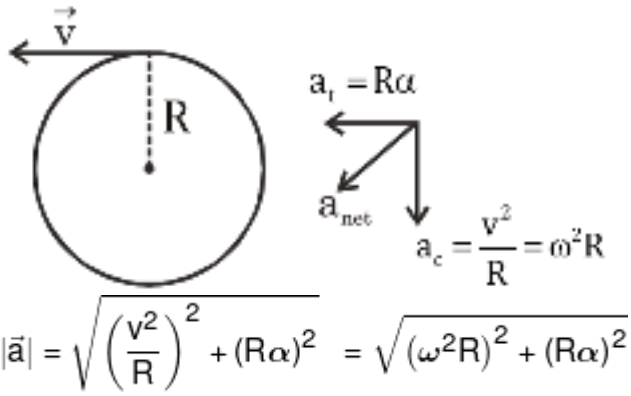
Final answer : 4

45) **Question Explanation:** The problem asks to identify the true statement(s) for a particle moving in a circle with constant angular acceleration.

Concept: This question is based on acceleration, velocity vector, circular motion.

Solution:

ω changes with time due to constant angular acceleration, the a_c is not constant.



$$|\vec{a}| = \sqrt{\left(\frac{v^2}{R}\right)^2 + (R\alpha)^2} = \sqrt{(\omega^2 R)^2 + (R\alpha)^2}$$

Since, a_c is not constant the magnitude of acceleration is also not constant. Therefore statement-I is false.
The tangential component is along the tangent but the centripetal component is directed towards the centre of the circle.
Therefore, the resultant acceleration vector is generally not along the tangent to the circle unless the centripetal acceleration is zero.
In circular motion, the instantaneous velocity vector is always tangential to the circular path at any given point.
The acceleration vector is not always perpendicular to the velocity vector.

Final answer: The correct option is (3).

CHEMISTRY

46) **Explaining :** Identify the balanced chemical reaction from the given rate expression
Concept : This question is based on rate of reaction in terms of stoichiometry
Solution : Minus signs are for reactants and positive signs for products and dividing numbers are the coefficients.
Final Answer : (2)

47) **Explaining :** Find overall order of the given complex reaction
Concept : This question is based on rate of complex reaction
Solution :
$r = k[A][B_2]$
and $k_c = \frac{[A][A]}{A_2}$ or $[A] = k_c^{1/2}[A_2]^{1/2}$
Thus $r = k \cdot k_c^{1/2}[A_2]^{1/2}[B_2]$
$= k' [A_2]^{1/2}[B]_2$
or order is $\frac{3}{2}$
Final Answer : (3)

48) **Explaining :** Find unit of rate constant from the given rate expression
Concept : This question is based on unit of rate constant
Solution :
Total order $= 1 + \frac{2}{3} = \frac{5}{3}$

so unit of rate constant = $(mol)^{1-\frac{5}{3}}(lit)^{\frac{5}{3}-1}sec^{-1}$
Final Answer : (1)

49) **Explaining :** For a first order reaction, if the time taken for completion of 50% of the reaction is t second, the time required for completion of 99.9% of the reaction is :
Concept : This question is based on first order reactions
Solution :
$t = \frac{0.693}{k}$ for first order reaction
$k = \frac{2.303}{t}\log\left(\frac{a}{a-x}\right)$
$\frac{0.693}{t_{1/2}} = \frac{2.303}{t_{99.9\%}}\log\left(\frac{100}{100-99.9}\right)$
$t_{99.9\%} = \frac{6.903}{0.693} \approx 10\,t$
Final Answer : (2)

50) **Explaining :** For the first order reaction with rate constant k, which expression gives the half-life period (Initial concentration = a) :
Concept : This question is based on half life of first order reaction
Solution : $t_{1/2} = \frac{0.693}{k} = \frac{2.303\log 2}{k} = \frac{\ln 2}{k}$
Final Answer : (1)

51) **Explaining :** Half-lives of first order and zeroth order reactions are same. Ratio of rates at the start of reaction is :
Concept : This question is based on half lives of zero and first order reaction
Solution :
$t_{50}\ (zeroth) = \frac{a}{2k_0}$
$t_{50}\ (first) = \frac{0.693}{k_1}$
$\frac{a}{2k_0} = \frac{0.693}{k_1}$
$\frac{k_1}{k_0} = \frac{2 \times 0.693}{a}$
$\frac{\left(\frac{dx}{dt}\right)_1}{\left(\frac{dx}{dt}\right)_0} = \frac{k_1 a}{k_0}$
$= 2 \times 0.693$
Final Answer : (2)

52) **Explaining :** The rate constant is numerically the same for three reaction of first, second and third order respectively which one is true for the three of reaction, if concentration of reactant is greater than 1M?
Concept : This question is based on rate law
Solution :
$r_1 = k[A]$ $r_2 = k[A]^2$ $r_3 = k[A]^3$

if [A] > 1 then $r_3 > r_2 > r_1$
Final Answer : (3)

53) **Explaining :** An elementary reaction between A and B is a second order reaction. Which of the following rate equations must be correct ?
Concept : This question is based on rate law of an elementary reaction
Solution :
Elementary reaction means a single step reaction
A + B → C (experimental reaction)
Final Answer : (4)

54) **Explaining :** If the rate of gaseous reactions is independent of pressure, the order of reaction is :
Concept : This question is based on order of reaction
Solution :
For zero order reaction, r = k (independent of the concentration and pressure of reactants)
Final Answer : (1)

55) **Explaining :** The rate constant of a reaction becomes equal to the pre-exponential factor when :
Concept : This question is based on Arrhenius equation
Solution :
Arrhenius equation
$$\log k = \log A - \frac{E_a}{2.303RT}$$
when $T \to \infty$ then $\log k = \log A$
or $k = A$
Final Answer : (1)

56) **Explaining :** Rate constant depends on
Concept : This question is based on rate constant
Solution : Rate constant depends on presence of catalyst and temperature
Final Answer : (1)

57) **Explaining :** Match the column-I with Column-II :
Concept : This question is based on prediction of order from graph
Solution :
Zero order ; $[A]_t = [A]_0 - kt$
First order ; $r = k[A]$
Second order ; $\frac{1}{[A]_t} = \frac{1}{[A]_0} + kt$
Third order ; $r = k[A]^3$
Final Answer : (2)

58) **Explaining :** Match the half-lives in Column-I with the orders in Column-II :
Concept : This question is based on half life
Solution :

Zero order ; $t_{1/2} = \frac{a_0}{2k}$

First order ; $t_{1/2} = \frac{0.693}{k}$

Second order ; $t_{1/2} = \frac{1}{a_0 k}$

Final Answer : (1)

59) **Explaining :** Calculate the rate constant of given reaction

Concept : This question is based on calculation of rate constant

Solution :

Ratio of equation $\frac{(1)}{(2)}$ $\quad \frac{1}{2} = \left[\frac{1}{2}\right]^m$

$\therefore$ m = 1

Ratio of equation $\frac{(2)}{(3)}$ $\quad \frac{1}{2} = \left(\frac{1}{2}\right)^n$

n = 1

So rate equation $r = k[A_2]^1 [B_2]^1$

Order = 1 + 1 = 2

Final Answer : (1)

60) **Explaining :** The process of reaction can be described by which of the following graphs?

Concept : This question is based on graph of exothermic reactions

Solution :

For an exothermic reaction, the potential energy of the products (AB) must be lower than the potential energy of the reactants (A + B).

Since, the reaction involves an intermediate (X), the graph must have two peaks. The first step, being fast needs less activation energy. The second step being slow, needs more activation energy.

$\therefore$ second peak will be higher than the first peak.

The correct graph is (3)

Final Answer : (3)

61) **Explaining :** Match the Column-I with Column-II :

Concept : This question is based on Van't hoff factor

Solution :

For dissociation, i > 1

(A) $Ca_3(PO_4)_2 \rightarrow 3Ca^{2+} + 2PO_4^{3-}$; i = 1 + (n-1) α

assuming 100% dissociation i = n = 5

(B) $Al_2(SO_4)_3 \rightarrow 2Al^{3+} + 3SO_4^{2-}$

i = 5

(C) Urea $\rightarrow$ i = 1

(D) CH_3COOH in benzene associates to form dimer $\therefore$ i < 1

Final Answer : (2)

62) **Explaining :** Freezing point of an aqueous solution is –0.186°C.

Concept : This question is based on elevation in boiling point

Solution :

$$\Delta T_f = \frac{1000 \cdot K_f \cdot w}{W \cdot m}$$

and $\Delta T_b = \frac{1000 \cdot K_b \cdot w}{W \cdot m}$

$\therefore \frac{\Delta T_f}{\Delta T_b} = \frac{K_f}{K_b}$ or $\frac{0.186}{\Delta T_b} = \frac{1.86}{0.512}$

$\therefore \Delta T_b = 0.0512°C$

Final Answer : (2)

63) **Explaining :** The van't Hoff factor (i) for a dilute aqueous solution of the strong electrolyte barium hydroxide is:

Concept : This question is based on Van't Hoff factor for dissociation

Solution : For $Ba(OH)_2$, i = n = 3

Final Answer : (4)

64) **Explaining :** A 5% (w/v) solution of cane sugar (molecular weight = 342) is isotonic with 1% (w/v) solution of a non-electrolyte substance X. The molecular weight of X is:

Concept : This question is based on isotonic solutions

Solution :

For Isotonic $C_1 = C_2$ (at same temperature)

$$\frac{w_1}{M_{w_1} \times V_1} = \frac{w_2}{M_{w_2} \times V_2}$$

$$\frac{5}{342 \times 100} = \frac{1}{M_{w_2} \times 100}$$

$M_{w_2} = 68.4$

Final Answer : (2)

65) **Explaining :** 0.5 M aqueous solution of glucose is isotonic with:

Concept : This question is based on isotonic solutions

Solution : For isotonic solutions ; $\pi_1 = \pi_2$ or $ic_1 = ic_2$ (at same temperature)

Final Answer : (3)

66) **Explaining :** Which of the following is dependent on temperature?

Concept : This question is based on concentration terms

Solution : Molarity is volume-dependent so it changes with temperature

Final Answer : (1)

67) **Explaining :** Normality of 10% (w/v) H_2SO_4 solution nearly :

Concept : This question is based on normality

Solution :

$$N = \frac{w \times 1000}{E_A \times V_{mL}} = \frac{10 \times 1000}{49 \times 100}$$

$N \simeq 2$

Final Answer : (4)

68) **Explaining :** 5 N H_2SO_4 was diluted from 1 litre to 10 litres. Normality of the solution

obtained is:
Concept : This question is based on concept of dilution
Solution :
$N_1V_1 = N_2V_2$
$5 \times 1 = N_2 \times 10$
$N_2 = 0.5$ N
Final Answer : (4)

69) **Explaining :** OR will be equal to :
Concept : This question is based on Raoult's Law
Solution : $P_T = P_A + P_B$ or OR = OQ + OP
Final Answer : (3)

70) **Explaining :** Which one is not equal to zero for an ideal solution:
Concept : This question is based on ideal solutions.
Solution : $\Delta S_{mix} > 0$
Final Answer : (1)

71) **Explaining :** For an ideal solution of two components A and B, which of the following is true?
Concept : This question is based on ideal solutions
Solution :
For an ideal solution:
$\Delta H_{mix} = 0$; $\Delta S_{mix} > 0$
$\Delta V_{mix} = 0$; $\Delta G_{mix} < 0$
A-A interactions $\simeq$ B-B interactions $\simeq$ A-B interactions
Final Answer : (3)

72) **Explaining :** The rate constant of most of the reactions increases with increase in temperature. According to the Arrhenius equation, $k = Ae^{-E_a/RT}$. The curve of rate constant 'k' against temperature 'T' will be :
Concept : This question is based on Arrhenius equation
Solution : $k = Ae^{-E_a/RT}$
Final Answer : (4)

73) **Explaining :** The rate constant of a reaction is 2.1×10^{-2} mol^{-2} $litre^2$ min^{-1}. The order of reaction is:
Concept : This question is based on unit of rate constant
Solution :
Unit of rate constant
$k = mol^{1-(m+n)}$ $lit^{(m+n-1)}$ $time^{-1}$
Final Answer : (3)

74) **Explaining :** Find the rate law expressions
Concept : This question is based on rate of complex reactions
Solution :

From slow step
$r = k [O] [O_3]$
[O] is intermediate so it will be replaced by using another step (fast step)

$$K_c = \frac{[O_2][O]}{[O_3]} \quad \text{or} \quad [O] = K_c \frac{[O_3]}{[O_2]}$$

then $$r = kK_c \frac{[O_3]^2}{[O_2]}$$

Final Answer : (4)

75) **Explaining :** The temperature coefficient for most of the reactions lies between:
Concept : This question is based on temperature coefficient
Solution : Temperature coefficient has a value between 2 and 3 for most reactions
Final Answer : (2)

76) **Explaining :** The reaction $2NO(g) + O_2(g) \rightleftharpoons 2NO_2(g)$ is of first order. If volume of reaction vessel is reduced to $\frac{1}{3}$, the rate of reaction would be :
Concept : This question is based on rate law
Solution :
$r = k[NO_2]^1$

$$r = k\left(\frac{n_{NO_2}}{V}\right)^1 \qquad r' = k\left(\frac{n_{NO_2}}{V/3}\right)^1$$

$\frac{r}{r'} = \frac{1}{3}$ ⇒ r' = 3r
3 times
Final Answer : (3)

77) **Explaining :** If in a 1st order reaction 75% of reaction complete in 4 hrs. Then how much time is required to complete 87.5% of reaction?
Concept : This question is based on fractional lives of first order reaction.
Solution :
For 75% $$4 = \frac{2.303}{k} \log \frac{100}{25}$$
For 87.5% $$t = \frac{2.303}{k} \log \frac{100}{12.5}$$
on there ratio $$\frac{4}{t} = \frac{\log 4}{\log 8} = \frac{0.6}{0.9}$$
$$t = \frac{4 \times 0.9}{0.6} = 6 \text{ hrs}$$
Final Answer : (2)

78) **Explaining :** When initial concentration of a reactant is doubled in a reaction, its half-life period is not affected. The order of the reaction is :
Concept : This question is based on determination of order of reaction
Solution : For first order reaction ; $t_{1/2} = \frac{0.693}{k} \quad (t_{1/2} \propto a^0)$

Final Answer : (1)

79) **Explaining :** Slope of the straight line obtained by plotting $\log_{10} k$ against 1/T represents which term?
Concept : This question is based on Arrhenius equation
Solution :
$k = Ae^{-E_a/RT}$
or $\log k = \log A - \frac{E_a}{2.303RT}$
$y = c + m x$
$\therefore$ slope $= = \frac{-E_a}{2.303R}$
Final Answer : (3)

80) **Explaining :** If 18 g of glucose is present in 1000 g of solvent, the solution is said to be:
Concept : This question is based on molality
Solution : $m = \frac{w_A \times 1000}{m_A \times w_B} = \frac{18 \times 1000}{180 \times 1000} = 0.1 \text{ molal}$
Final Answer : (4)

81) **Explaining :** The depression in f.pt. of 0.01 m aqueous solution of urea, sodium chloride and sodium sulphate is in the ratio:
Concept : This question is based on depression in freezing point
Solution :
Net Molarity or Molality = m × no. of particles

Urea	:	NaCl	:	Na_2SO_4
0.01×1		0.01×2		0.01×3
1	:	2	:	3

Final Answer : (3)

82) **Explaining :** Vapour pressure of a solution containing non-volatile solute is:
Concept : This question is based on vapour pressure
Solution :
V.P. of solution = V.P. of solvent in the solution
which is directly proportional to mole fraction of solvent in the solution.
Final Answer : (2)

83) **Explaining :** Value of Henry's constant (K_H)__.
Concept : This question is based on Henry's law
Solution : Value of Henry's constant depends on nature of gas and temperature. K_H increases with increase in temperature.
Final Answer : (2)

84) **Explaining :** If molality of the dilute solutions is doubled, the value of molal depression constant (K_f) will be:
Concept : This question is based on molal depression constant

Solution : k_f depends only on nature of solvent.
Final Answer : (3)

85) **Explaining :** The mixture of two liquids boils without changing its composition is called:
Concept : This question is based on Azeotropic mixtures
Solution :
Azeotropes are binary liquid mixtures having same composition in liquid and vapour phase and boil at a constant temperature
Final Answer : (3)

86) **Explaining :** What is the freezing point of a solution that contains 10.0 g of a glucose ($C_6H_{12}O_6$) in 100 g of H_2O ?
Concept : This question is based on depression in freezing point
Solution :

$$\Delta T_f = \frac{K_f \times W_A \times 1000}{M_A \times W_B} = \frac{1.86 \times 10 \times 1000}{180 \times 100}$$

$T_f - T' = 1.03$
$0 - T' = 1.03$
$T' = -1.03$
Final Answer : (4)

87) **Explaining :** Which of the following is not a non-electrolyte?
Concept : This question is based on electrolytes
Solution : Acetic acid is a weak electrolyte, so it is not a non-electrolyte.
Final Answer : (1)

88) **Explaining :** Pure water can be obtained from sea water by:
Concept : This question is based on reverse osmosis
Solution : Reverse osmosis is used in desalination of sea water.
Final Answer : (3)

89) **Explaining :** What is the van't Hoff factor for sodium phosphate?
Concept : This question is based on Van't Hoff factor
Solution : Na_3PO_4 i = n = no. of ions = 4
Final Answer : (4)

90) **Explaining :** Van't Hoff factor more than unity indicates that the solute in solution has:
Concept : This question is based on Van't Hoff factor
Solution : In case of dissociation, value of Van't Hoff factor (i) is greater than unity.
Final Answer : (2)

BIOLOGY

91) **Solution :** All the given statements are true for living organisms.

Final Answer : (3)

92) **Solution :** Extrinsic growth is not considered as defining feature of living being organisms.
Final Answer : (1)

93) **Solution :** In both plants and animals the most convenient method of nomenclature system is binomial nomenclature
Final Answer : (2)

94) **Solution :** Infertile human couples and mules cannot reproduce.
Final Answer : (4)

95) **Solution**: Statements A, B, and D are correct. Biological names are in Latin. Animals, mammals, and dogs are valid taxa at different levels. The biological species concept applies only to sexually reproducing organisms, not bacteria.
Final Answer: (1)

96) **Solution :** Species is the smallest fundamental unit of taxonomy.
Final Answer : (3)

97) **Solution :** The **biological concept of species** is based on the ability of organisms to interbreed and produce fertile offspring. This concept emphasizes reproductive isolation, meaning members of a species can reproduce among themselves but not with members of other species.
Final Answer : (2)

98) **Explain Question:** The question asks the taxonomic similarity between *Panthera* and *Solanum*.
Concept: Genus
Solution:
Panthera → a genus of animals (lion, tiger, leopard).
Solanum → a genus of plants (potato, brinjal, tomato).Both represent the same taxonomic rank: genus.
Final answer: (3)

99) **Solution:** (A) Lower taxa (species, genus) show more similarities — True.
(B) Families are described using both vegetative and reproductive features — True (though reproductive features are more important).
(C) Felidae = cat family and Canidae = dog family — True.
(D) Higher taxa have fewer similarities, making relationships harder to determine — True.
All statements are correct.
Final Answer : (4)

100) **Solution :** Genus of tiger, leopard and lion is *Panthera*.

Genus of Mango is *Mangifera*.
Genus of Wheat is *Triticum*
Genus of Rice is *Oryza*
Genus of Brinjal, Makoi and Potato is *Solanum*
Genus of Housefly is *Musca*
Genus of Cat is *Felis*
Final Answer : (2)

101) **Solution :** a-v, b-iv, c-ii, d-i, e-iii

	Column-I		**Column-II**
(a)	Family	(v)	Solanaceae
(b)	Kingdom	(iv)	Plantae
(c)	Order	(ii)	Polymoniales
(d)	Specific epithet	(i)	*tuberosum*
(e)	Genus	(iii)	*Solanum*

Final Answer : (2)

102) **Solution :** Family of mango is Anacardiaceae
Final Answer : (1)

103) **Solution :** Potato, Brinjal, Makoi, *Petunia* and *Datura* have some common characters. So they are placed in the same family
Final Answer : (4)

104) **Solution :** Cats belong to Felidae family whereas dogs belong to Canidae family
Final Answer : (1)

105) **Solution :** The word systematics is derived from Latin word '*Systema*' which means systematic arrangement of organisms.
Final Answer : (1)

106) **Solution:** *Volvox* – Correct
Volvox is a green alga.
It forms a motile colony — a spherical colony of thousands of cells.
The colony can move using flagella present on individual cells that coordinate together.
This is a classic example of a motile colonial alga.
Final Answer: (1)

107) **Solution :** Algae are mainly classified on the basis of types of pigment present in cell.
Final Answer : (4)

108) **Solution :** *Spirogyra* shows isogamy with non-flagellated gametes.
Final Answer : (4)

109) **Solution :** Sex organs of algae are usualy unicellular and non Jacketed.
Final Answer : (2)

110) **Solution :**
(a) *Volvox* is colonial algae.
(d) *Porphyra, Laminaria* and *Sargassum* are used as food.
Final Answer : (2)

111) **Solution :** A protonema is a creeping, green, filamentous and gametophytic structure produced in bryophytes.
Final Answer : (3)

112) **Solution :** Bryophytes are called amphibians of plant kingdom because they need water for fertilization.
Final Answer : (4)

113) **Solution :** From evolutionary point of view, retention of the female gametophyte with developing young embryo on the parent sporophyte for some time, is first observed in pteridophytes.
Final Answer : (3)

114) **Solution :** Genera like *Selaginella* and *Salvinia* produce two kinds of spores. Such plants are known as Heterosporous.
Final Answer : (4)

115) **Solution :** A fern is different from moss in terms of having vascular tissue.
Final Answer : (2)

116) **Solution :** Floridean starch is reserve food in rhodophyceae.
Final Answer : (3)

117) **Solution :** Stomata are not absent in *Pinus* leaf instead sunken stomata is present to present transpiration.
Final Answer : (4)

118) **Solution :** "They occur at great depth in oceans where little light penetrates." Above character best suited to red algae.
Final Answer : (1)

119) **Solution :** Prothallus is small, inconspicuous, multicellular, free-living gametophyte.

Final Answer : (2)

120) **Solution :** In *Cycas* female plant does not have a compact cone. Instead, it produces a loose cluster of Megasporophylls at the apex of the stem. These megasporophylls are leaf-like and remain separate (lax), rather than being tightly packed into a cone structure.
Final Answer : (4)

121) **Solution :** Bryophyta have small sporophyte phase which is generally dependent on gametophyte.
Final Answer : (4)

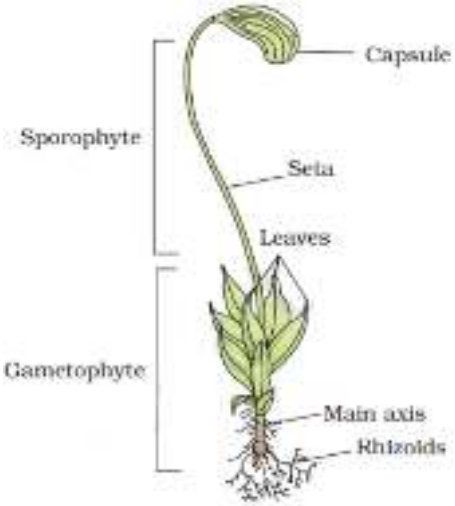
122) **Solution :** Zygotic meiosis is not true for bryophytes, instead sporic meiosis takes place.
Final Answer : (3)

123) **Solution :** The main plant body of bryophytes is **haploid**. It produces **gametes**, hence is called a **gametophyte**.
Final Answer : (2)

124) **Solution :** "Gemmae" are multicellular green structures for vegetative propagation. These are found in thallus of *Marchantia.*
Final Answer : (2)

125)

Explain Question : Question asking correct labelling part about the given diagram
Concept : This question is based on Bryophyta
Solution : Given diagram represent gametophyte and sporophyte of *Funaria* and labelled part are :
A-Capsule, B-Seta, C-leaves, D- Rhizoids



Final Answer : (4)

126) **Solution :** Vegetative reproduction in mosses takes place by Fragmentation and Budding in secondary protonema.

Final Answer : (4)

127) **Solution :** Peat moss (*Sphagnum*) is a moss.
Final Answer : (4)

128) **Solution :** Protonema is formed in life cycle of moss, between spore and main plant.
Final Answer : (4)

129) **Solution :** In moss reduction division takes place in capsule.
Final Answer : (2)

130) **Solution :** Branched chain lipids occur in the cell membranes of archaebacteria.
Final Answer : (1)

131) **Solution :** Smallest known moneran lacking cell wall are mycoplasma.
Final Answer : (2)

132) **Solution :** 'Contagium vivum fluidum' (i.e., living fluid infectant) term has given by Beijerinck.
Final Answer : (3)

133) **Solution :** a – 4, b – 3, c – 2, d – 1
a. Plantae → 4. Algae
b. Fungi → 3. Phycomycetes
c. Protista → 2. Euglenoids
d. Monera → 1. Archaebacteria
Final Answer : (1)

134) **Explain Question:** The question asks which statements about lichens are correct or incorrect.
Concept: Kingdom Fungi, Lichens.
Solution: Lichens are made up of algal component (phycobiont) and fungal component (mycobiont).
Final Answer: (1)

135) **Solution :** Nitrogen fixation occurs in specialised cells called heterocyst.
Final Answer : (1)

136) **Solution :** Chemosynthetic autotropic bacteria oxidise various inorganic substances such as nitrates, nitrites & ammonia and use the released energy for ATP production.
Final Answer : (3)

137) **Solution :** Infoldings of plasma membrane in bacteria are called mesosomes.

Final Answer : (4)

138) **Solution:** *Methanogens* are a group of archaebacteria that produce methane as a by-product of their metabolic processes. They are utilized in Gober gas (biogas) plants for the production of methane gas from organic waste. These bacteria thrive in anaerobic conditions and help break down organic material, producing methane gas as a renewable energy source.
Final answer: Methanogens

139) **Solution:** ***Rhizobium* -** Hetero trophic - Lives symbiotically in legume roots (Fixes N_2).
***Lactobacillus* -** Heterotrophic - Used in curd production, breakdown lactose. **Final Answer: (3)**

140) **Solution :** Organism of which kingdom feed like animals and perform photosynthesis like plants Protista.
Final Answer : (2)

141) **Solution :** Armoured cell wall and biflagellated cells are characteristic of Pyrrophyta.
Final Answer : (2)

142) **Explanation:**
→ Certain dinoflagellates, such as Noctiluca and Gonyaulax, exhibit bioluminescence, emitting a blue-green glow in seawater at night.
→ This occurs due to the luciferin-luciferase reaction, where the enzyme luciferase catalyzes the oxidation of luciferin, releasing light.
→ This glowing effect makes them look like fire in the water, hence the name "fire algae."
Final Answer: (4)

143) **Solution :** *Gonyaulax* is classified under dinoflagellates and diatoms under chrysophytes.
Final Answer : (1)

144) **Solution :** Both statement are correct.
Final Answer : (3)

145) **Solution :** Chlorophyll 'b' is present in Euglenoid but it is absent in in dinoflagellates and diatoms.
Final Answer : (3)

146) **Solution :** They are photosynthetic in presence of sunlight, when deprived of sunlight, they behave like heterotrophs by predating on small organisms.
Final Answer : (2)

147) **Solution :** Phycomycetes produces Aplanospores, Zoospores and Zygospores.
Final Answer : (1)

148) **Solution:**
Albugo is parasitic fungus-like organism.
It causes white rust disease on mustard and other cruciferous plants.
Final Answer : (1)

149) **Solution:** *Rhizopus* is a member of Zygomycetes, not Basidiomycetes. **Final Answer : (4)**

150) **Solution :** a–i, b–iii, c–ii
Final Answer : (2)

151) **Solution :** Poriferans possess a canal system; Coelenterates show radial symmetry; Platyhelminthes have flame cells; Annelids exhibit metameric segmentation.
Final Answer : (1)

152) **Solution :** A-ii, B-iii, C-i, D-iv
Sycon is a sponge (Porifera), Hydra belongs to Coelenterata, Ascaris is a roundworm (Aschelminthes), and Nereis is an annelid.
Final Answer : (3)

153) **Solution :** A-ii, B-i, C-iii, D-iv
Cnidoblasts are unique to Coelenterata. Comb plates are characteristic of Ctenophora. Pseudocoelom is present in Aschelminthes. Setae occur in Annelida.
Final Answer : (4)

154) **Solution :** A-ii, B-iv, C-i, D-iii
Porifera have cellular organisation, Coelenterata have tissue-level organisation, Platyhelminthes are acoelomates, and Annelids possess a true coelom.
Final Answer : (3)

155) **Solution :** Statement D is incorrect because Poriferans show only cellular level of organisation.
Final Answer : (2)

156) **Solution :** Ctenophores possess colloblasts, not cnidoblasts.
Final Answer : (1)

157) **Solution :** Platyhelminthes are acoelomates, not pseudocoelomates.
Final Answer : (3)

158) **Solution :** Earthworm is monoecious (hermaphrodite), not dioecious.
Final Answer : (3)

159) **Solution :** Flame cells occur in cephalochordata for excretion.
Final Answer : (1)

160) **Solution :** Sponges are multicellular organisms with cellular-level organisation.
Final Answer : (2)

161) **Solution :** The dorsoventrally flattened body gives these animals the name 'flatworms'.
Final Answer : (1)

162) **Solution :** Comb plates help in locomotion, whereas bioluminescence is a separate characteristic.
Final Answer : (2)

163) **Solution :** Choanocytes (collar cells) are characteristic cells of sponges.
Final Answer : (4)

164) **Solution :** Aschelminthes (Nematoda) have a cylindrical, unsegmented body and are commonly called roundworms.
Final Answer : (2)

165) **Solution :** Earthworm, an annelid, shows true metameric segmentation, which improves movement and body organisation.
Final Answer : (3)

166) **Solution :** Jointed appendages are characteristic of Arthropoda. Molluscs possess a muscular foot. Echinoderms have a water vascular system, while Hemichordates possess a proboscis.
Final Answer : (3)

167) **Solution :** Pila is a mollusc, Starfish belongs to Echinodermata, Cockroach is an arthropod, and Balanoglossus belongs to Hemichordata. **Final Answer : (4)**

168) **Solution :** Cartilaginous fishes have cartilage, bony fishes possess a swim bladder, amphibians have moist skin, and birds possess pneumatic bones.
Final Answer : (1)

169) **Solution :** Herdmania is a urochordate, Branchiostoma is a cephalochordate, Petromyzon is a cyclostome, and Whale is a mammal.
Final Answer : (2)

170) **Solution :** All statements are correct.
Final Answer : (4)

171) **Solution :** Muscular foot is found in Mollusca, not Echinodermata.
Final Answer : (1)

172) **Solution :** Chordates possess a dorsal hollow nerve cord, not a ventral solid nerve cord.
Final Answer : (2)

173) **Solution :** Both Chondrichthyes and Osteichthyes possess a two-chambered heart.
Final Answer : (3)

174) **Solution :** Feathers are characteristic of birds, not amphibians.
Final Answer : (3)

175) **Solution :** Mammary glands are the defining feature of mammals and provide nourishment to offspring.
Final Answer : (1)

176) **Solution :** Amphibians generally have aquatic larvae and terrestrial or semi-aquatic adults.
Final Answer : (1)

177) **Solution :** Air-filled (pneumatic) bones make the skeleton lighter, helping birds fly efficiently.
Final Answer : (1)

178) **Solution :** Petromyzon (lamprey) is a jawless vertebrate belonging to Cyclostomata.
Final Answer : (2)

179) **Solution :** Cycloid and ctenoid scales are characteristic of cartilaginous fishes such as sharks and rays.
Final Answer : (1)

180) **Solution :** Whales breathe through lungs, give birth to young ones, and feed them with milk, making them mammals.
Final Answer : (1)